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Selective modulation of protein-protein interactions

Abstract

The present disclosure provides methods to identify peptides and small molecule moieties that are able to modulate protein-protein interactions (PPIs). Some moieties can disrupt specific PPIs within a complex, or disrupt variant-specific PPIs. Some moieties can alternatively bridge between two proteins in a protein-specific or a variant-specific manner. The methods described enable generation of compounds able to modulate PPI networks within cells with implications for drug development for pathological conditions.

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Background/Summary

CROSS-REFERENCE (1) This application is a continuation application of U.S. Non-Provisional application Ser. No. 15/931,295, filed May 13, 2020, which is a continuation application of International Application No. PCT/US2018/061292, filed Nov. 15, 2018, which claims the benefit

of U.S. Provisional Application No. 62/587,269 titled SELECTIVE DISRUPTION OF PROTEIN-PROTEIN INTERACTIONS, filed on Nov. 16, 2017 and U.S. Provisional Application No. 62/590,147 titled CYCLIC AND BICYCLIC PEPTIDES AND METHODS OF MAKING AND USING THEREOF, filed on Nov. 22, 2017, which are incorporated herein by reference.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML file, created on Oct. 31, 2023, is named 50607-703_302_SL.xml and is 96,288 bytes in size.

BACKGROUND

(2) Disruption of protein-protein interactions in a precise manner can be a key for controlling cellular functions. Many pathological conditions are characterized by aberrant functions of cellular pathways, either because of precocious protein complex formation or the incorporation of malfunctional variants. Thus, compounds that can specifically and precisely prevent the formation of such protein complexes or the malfunction of faulty variants could be beneficial to treating various ailments. The selective disruption of precise protein-protein interactions is difficult to achieve using the traditional enzyme active-site/inhibitor-based drug development scheme. Accordingly, there is a need for development of methods and compositions that target protein-protein interactions in precise and selective ways.

SUMMARY

(3) Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively disrupts an interaction between a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell, wherein a sequence of gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein, wherein a positive selection reporter is disposed within the host and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein, and wherein, in the absence of the molecule, the interaction between the first test protein and the second test protein causes the gene activating moiety to activate expression of the death agent, while the interaction between the second test protein and the third test protein causes the gene activating moiety to activate the expression of the positive selection reporter. In some embodiments, the molecule from the library is delivered exogenously. In some embodiments, the host cell comprises more than one sequence for expressing a positive control reporter that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises more than one sequence for expressing a death agent that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein; a plasmid DNA encoding the death agent; and a plasmid DNA encoding a positive selection reporter. In some embodiments, the first test protein is a variation of KRAS. In some embodiments, the third test protein is KRAS. In some embodiments, the second test protein is c-Raf. In some embodiments, the first test protein is YAP or TAZ. In some embodiments, the third test protein is VGLL4. In some embodiments, the second test protein is TEAD. In some embodiments, the DNA binding moiety is derived from LexA, cI, Gli-1, YY1, Glucocorticoid receptor, TetR, or Ume6. In some embodiments, the gene activating moiety is derived from VP16, GAL4, NF- κ B, B42, BP64, VP64, or p65. In some embodiments, the death agent is an overexpressed product of genetic element selected from DNA or RNA. In some embodiments, the genetic element is a Growth Inhibitory

(GIN) sequence such as GIN11. In some embodiments, the death agent is a ribosomally encoded xenobiotic agent, a ribosomally encoded poison, a ribosomally encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, or any combination thereof. In some embodiments, the death agent is Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phallacidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, or any combination thereof. In some embodiments, the host cell is a fungus or bacteria. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*. In some embodiments, the molecule is small molecule. In some embodiments, the small molecule is peptidomimetic. In some embodiments, the molecule is peptide or protein. In some embodiments, the peptide or protein is derived from naturally occurring protein product. In certain embodiments, the peptide or protein is synthesized protein product. In some embodiments, the peptide or protein is product of recombinant genes. In some embodiments, the molecule is a peptide or protein expressed from test DNA molecule inserted into the host cell, wherein the test DNA molecule comprises DNA sequences that encodes polypeptides, forming the library. In some embodiments, the library comprises polypeptides 60 or fewer amino acids in length. In some embodiments, the DNA sequence encodes a 3'UTR of mRNA. In some embodiments, the 3'UTR is the 3'UTR of sORF1. In some embodiments, the polypeptides comprise a common N-terminal sequence of Methionine-Valine-Asparagine. In some embodiments, the polypeptides in the library are processed into cyclic or bicyclic peptides in the host cell.

(4) Disclosed herein, in certain embodiments, is a plasmid vector, comprising the components of PLASMID 1, or any combination of the components of PLASMID 1. In some embodiments, the plasmid vector comprises a DNA sequence encoding a first polypeptide inserted in frame with Gal4-DNA binding domain ("DBD"), a DNA sequence encoding a second polypeptide inserted in frame with LexA-DBD, and a DNA sequence encoding a third polypeptide inserted in frame with Dof1-AD. In certain embodiments, a host cell comprises the plasmid vectors.

(5) Disclosed herein, in certain embodiments, is a library of plasmid vectors, each plasmid vector comprising: a DNA sequence encoding a different peptide sequence operably linked to a first switchable promoter; a DNA sequence encoding a death agent under control of a second switchable promoter; and a DNA sequence encoding a positive selection reporter under control of a third switchable promoter. In some embodiments, the different peptide sequence encodes a common N-terminal stabilization sequence. In some embodiments, the DNA sequence encodes a mRNA sequence comprising a 3'UTR. In some embodiments, the different peptide sequence is 60 amino acids or fewer in length. In some embodiments, the different peptide sequences are random. In some embodiments, the different peptide sequences are pre-enriched for binding to a target. In some embodiments is a library of host cells, each comprises a library of the plasmid vectors.

(6) Disclosed herein, in certain embodiments, is a host cell configured to express: a first fusion protein comprising a DNA-binding moiety; a second fusion protein comprising a gene activating moiety; a third fusion protein comprising a different DNA-binding moiety; a death agent, wherein the expression of the death agent is under control of a promoter DNA sequence specific for one of the DNA-binding moiety; a positive selection reporter, wherein the expression of the positive reporter is under control of a promoter DNA sequence specific for the other DNA-binding moiety; and a polypeptide of 60 or fewer amino acids, wherein the polypeptide modulates an interaction between the first test protein and the second test protein; wherein the host cell optionally has a mutant background enabling uptake of small molecules; and wherein the host cell optionally has a

mutant background enabling increased transformation efficiency. In some embodiments, the polypeptide encodes an N-terminal sequence for peptide stabilization. In some embodiments, the polypeptide is an encoded product of an mRNA, wherein the mRNA comprises a 3'UTR. In some embodiments, the mRNA is an encoded product of a DNA molecule, wherein the DNA molecule is delivered into the host cell exogenously. In some embodiments, synthetic compound libraries can be tested. In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is animal, plant, a fungus, or bacteria. In some embodiments, the host cell is a haploid yeast cell. In some embodiments, the host cell is a diploid yeast cell. In some embodiments, the diploid yeast cell is produced by mating a first host cell comprising DNA sequences encoding the first chimeric gene, the second chimeric gene, and the third chimeric gene, to a second host cell comprising DNA sequences encoding the death agent, positive selection reporter, and the mRNA comprising a nucleotide sequence encoding a polypeptide. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*. In some embodiments is a kit, comprising: the plasmid vector and the library of plasmid vectors.

(7) Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively facilitates an interaction between a first test protein and a second test protein, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell such that the molecule forms a bridging interaction between the first test protein and the second test protein; wherein a sequence of a gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein; wherein a positive selection reporter is disposed within the host cell and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein; and wherein the first test protein and second test protein to form a functional transcription factor that activates expression of the death agent when the molecule from the library forms the bridging interaction. In some embodiments, the molecule from the library is delivered exogenously. In some embodiments, the host cell comprises more than one sequence for expressing a death agent that is activated by the promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises more than one sequence for expressing a positive control reporter that is activated by a promoter DNA sequence specific for a DNA binding moiety. In some embodiments, the host cell comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein; a plasmid DNA encoding the death agent; and a plasmid DNA encoding a positive selection reporter. In some embodiments, the DNA binding moiety is derived from LexA, cI, Gli-1, YY1, Glucocorticoid receptor, TetR, or Ume6. In some embodiments, the gene activating moiety is derived from VP16, Gal4, NF- κ B, B42, BP64, VP64, or p65. In some embodiments, the death agent is a genetic element wherein overexpression of genetic material results in growth inhibition of the host cell. In some embodiments, the death agent is an overexpressed product of DNA. In some embodiments, the death agent is an overexpressed product of RNA. In some embodiments, the sequence of the gene for expressing the death agent is a Growth Inhibitory (GIN) sequence such as GIN11. In some embodiments, the death agent is a ribosomally encoded xenobiotic agent, a ribosomally-encoded poison, a ribosomally-encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally-encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, or any combination thereof. In some embodiments, the death agent is Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE,

HopPtoF, HopPtoG, YopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phalloidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, or any combination thereof. In some embodiments, the first test protein is a variation of KRAS. In some embodiments, the third test protein is KRAS. In some embodiments, the second test protein is c-Raf. In some embodiments, the first test protein is YAP or TAZ. In some embodiments, the third test protein is VGLL4. In some embodiments, the second test protein is TEAD. In some embodiments, the molecule is small molecule. In some embodiments, the small molecule is peptidomimetic. In some embodiments, the molecule is peptide or protein. In some embodiments, the peptide or protein is derived from naturally occurring protein product. In some embodiments, the peptide or protein is synthesized protein product. In some embodiments, the peptide or protein is product of recombinant genes. In some embodiments, the peptide or protein is expressed product of test DNA molecule inserted into the host cell, wherein the test DNA molecule comprises of DNA sequences that encodes polypeptides, forming the library. In some embodiments, the library comprises of sixty or fewer amino acids. In some embodiments, the peptide or protein is a product of post-translational modification. In some embodiments, the post-translational modification includes cleavage. In some embodiments, the post-translational modification includes cyclization. In some embodiments, the post-translational modification includes bi-cyclization. In some embodiments, the cyclization comprises reacting with prolyl endopeptidase. In some embodiments, the cyclization comprises reacting with beta-lactamase. In some embodiments, the bicyclization comprises reacting with hydroxylase and dehydratase. In some embodiments, the bicyclization is formed by a tryptathionine bridge. In some embodiments, the post-translational modification includes methylation. In some embodiments, the methylation comprises reacting with N-methyltransferase. In some embodiments, the post-translational modification includes halogenation. In some embodiments, the post-translational modification includes glycosylation. In some embodiments, the post-translational modification includes acylation. In some embodiments, the post-translational modification includes phosphorylation. In some embodiments, the post-translational modification includes acetylation. In some embodiments, the test DNA molecule comprises of gene sequence expressing modifying enzyme. In some embodiments, the test DNA molecule comprises of a gene sequence expressing N-terminal sequence of methionine-valine-asparagine. In some embodiments, the test DNA molecule comprises of a gene sequence encoding a 3'UTR. In some embodiments, the 3'UTR is 3'UTR of sORF1. In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is animal, plant, a fungus, or bacteria. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*.

(8) Disclosed herein, in certain embodiments, is a method of identifying a molecule that selectively modulates a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a second fusion protein comprising the second test protein; delivering a first molecule to the host cell; modifying the first molecule while in the host cell via a modifying enzyme; and allowing the first molecule to modulate the interaction between the first test protein and the second test protein, wherein the first molecule is a product of an encoded DNA sequence, wherein the first molecule comprises a library and one or more modifying enzymes, and wherein the one or more modifying enzymes modify the library. In some embodiments, the first molecule is a small molecule. In some embodiments, the small molecule is peptidomimetic. In some embodiments, the first molecule is peptide or protein. In certain embodiments, the peptide or protein is derived from naturally occurring protein product. In some embodiments, the peptide or protein is synthesized protein product. In some embodiments, the first molecule is encoded in the host cell. In some embodiments, the first molecule is delivered exogenously. In some embodiments, the one or more modifying enzymes cause cleavage of the

library. In some embodiments, the one or more modifying enzymes cause cyclization of the library. In some embodiments, the one or more modifying enzymes cause bicyclization of the library. In some embodiments, the cyclization comprises reacting with prolyl endopeptidase. In some embodiments, the cyclization comprises reacting with beta-lactamase. In some embodiments, the bicyclization comprises reacting with hydroxylase and dehydratase. In some embodiments, the bicyclization comprises formation of a tryptathionine bridge. In some embodiments, the one or more modifying enzymes cause methylation. In some embodiments, the one or more modifying enzymes is a methyltransferase. In some embodiments, the one or more modifying enzyme is a halogenase. In some embodiments, the one or more modifying enzymes cause glycosylation. In some embodiments, the one or more modifying enzymes cause acylation. In some embodiments, the one or more modifying enzymes cause phosphorylation. In some embodiments, the one or more modifying enzymes cause acetylation. In some embodiments, the library comprises of sixty or fewer amino acids. In some embodiments, the first test protein is KRAS or a variation of KRAS. In some embodiments, the second test protein is c-Raf. In some embodiments, the first test protein is YAP, TAZ, or VGLL4. In some embodiments, the second test protein is TEAD. In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the host cell is animal, plant, a fungus, or bacteria. In some embodiments, the fungus is *Aspergillus*. In some embodiments, the fungus is *Pichia pastoris*. In some embodiments, the fungus is *S. cerevisiae*.

INCORPORATION BY REFERENCE

(9) All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features of the disclosure are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present disclosure will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the disclosure are utilized, and the accompanying drawings of which:

(2) FIGS. 1A and 1B illustrate a platform to identify a compound that specifically disrupts a protein-protein interaction. FIG. 1A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest (A-C). FIG. 1B illustrates a case where a peptide is able to disrupt the pairwise interaction of interest (A-C) without disrupting another interaction (A-B).

(3) FIGS. 2A and 2B illustrate a platform to identify a compound that disrupts a protein-protein interaction (A-B') in a variant-specific manner. FIG. 2A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest (A-B'). FIG. 2B illustrates a case where a peptide is able to disrupt the pairwise interaction of interest between one variant (B') and a protein (A) without disrupting the interaction between another variant (B) and the protein (A).

(4) FIGS. 3A and 3B exemplify an embodiment of a platform to identify a compound that bridges two proteins in a variant- or protein-specific manner. FIG. 3A shows a case where a peptide is able to bridge one variant (B) and a protein (A). FIG. 3B illustrates a case where a peptide is able to bridge the secondary variant (B') and a protein (A).

(5) FIGS. 4A and 4B show an embodiment of a platform to identify a compound that specifically disrupts a protein-protein interaction, where the platform includes two positive selection markers and a negative selection markers. In these figures, the compound may interact with the prey to disrupt the prey-bait interaction. FIG. 4A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest. FIG. 4B illustrates a case where a peptide is able to disrupt the Prey and BaitMut interaction by acting on the Prey without disrupting the interaction between BaitWT

and Prey.

(6) FIGS. 5A and 5B show an embodiment of a platform to identify a compound that specifically disrupts a protein-protein interaction in a system with two positive selections and a negative selection. In contrast with FIGS. 4A and 4B, in these figures, the compound may act on the bait (rather than the prey) to disrupt the prey-bait interaction. FIG. 5A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest. FIG. 5B illustrates a case where a peptide disrupts the Prey-BaitMut interaction by acting on the BaitMut without disrupting the interaction between BaitWT and Prey.

(7) FIG. 6 illustrates two platforms to identify a compound that specifically disrupts a protein-protein interaction in cell culture by selecting for nutritional reporter.

(8) FIG. 7 illustrates two platforms to identify a compound that specifically disrupts a protein-protein interaction in cell culture by selecting for cytotoxic reporter.

(9) FIG. 8 shows an embodiment of an integration plasmid that encodes two bait proteins, each with its own DNA binding domain (“DBD”), and a prey protein with an activation domain (“AD”).

(10) FIG. 9 shows an embodiment of a selection and library plasmid that encodes two death agents, both with the same DNA binding sequence, a positive selection agent driven by another DNA binding sequence and an inducible stabilized peptide library.

(11) FIG. 10 shows an embodiment of a confirmation plasmid that encodes two bait-prey fusion proteins, each with its own DBD.

(12) FIG. 11 illustrates a cyclization process of families of RiPP scaffolds that lead to the generation of N-to-C cyclized, backbone N-methylated macrocycles by the action of (1) prolyl oligopeptidases belonging to the PopB family and (2) N-methyltransferases belonging to the omphalotin methyltransferase family. The variable peptide library region is embedded within the primary sequence of the modifying enzyme. N- and C-term sequences refer to consensus binding and processing recognition elements. The N-terminus comprises the enzymatic N-methyltransferase domain, a linker region, and the processing enzymes binding sites.

(13) FIG. 12 illustrates a cyclization process of different families of RiPP scaffolds that lead to the generation of N-to-C cyclized, backbone N-methylated macrocycles by the action of a beta-lactamase (alanyl-alanine transpeptidase) and an N-methyltransferase. The variable peptide library region is embedded within the primary sequence of the modifying enzyme. N- and C-term sequences refer to consensus binding and processing recognition elements. The N-terminus comprises the enzymatic N-methyltransferase domain, a linker region, and the processing enzymes binding sites.

(14) FIG. 13 shows a gene organization within two sets exemplary loci from *Rhizopogon vinicolor* that encode for modifying enzymes that could be used in the schematic depicted in FIG. 12.

(15) FIG. 14 shows WebLogo alignments of a large variety of MSDIN family genes (toxin preproprotein sequences) identified in the genomes of *Amanita bisporigera* and *Amanita phalloides*. See Pulman, Jane A., et al. “Expansion and Diversification of the MSDIN Family of Cyclic Peptide Genes in the Poisonous Agarics *Amanita Phalloides* and *A. Bisporigera*.” *BMC Genomics*, vol. 17, no. 1, 15 Dec. 2016, p. 1038., doi: 10.1186/s12864-016-3378-7.

(16) FIG. 15 illustrates a cyclization process for members of the MSDIN family of RiPPs that leads to the generation of N-to-C cyclized macrocycles by the action of prolyl oligopeptidases belonging to the PopB family.

(17) FIG. 16 illustrates a bicyclization process for members of the MSDIN family of RiPPs that leads to the generation of N-to-C cyclized macrocycles and bicyclic macrocycles internally bridged via a tryptathionine bridge.

(18) FIG. 17 illustrates biochemical steps to create a tryptathionine bridge of a bicyclic ring.

DETAILED DESCRIPTION

(19) Two-hybrid screening can be used to identify and characterize protein-protein interactions. The two-hybrid system was initially developed using yeast as a host organism. However, bacterial

or animal cell two-hybrid systems can also be used to characterize protein-protein interactions. The present disclosure provides a system that can use a unified eukaryotic or prokaryotic two-hybrid system in which bait and prey expression plasmid is used in both organismal contexts. Additionally, an extensive series of leucine zipper fusion proteins of known affinities can be generated to compare the efficiency of interaction detection using both systems. The yeast system can produce a quantitative readout over a dynamic range. "Auto-activation" by baits can be less prevalent in the bacterial system. In addition, modified expression vectors disclosed herein can be used for expression of a protein of interest in both eukaryotes and prokaryotes.

(20) Three-hybrid systems rely on similar principles as two-hybrid systems, but involve an additional factor to bridge a protein-protein interaction to result in a gene expression readout.

(21) The present disclosure also provides a system for delivering molecules across the cell membrane. The cell membrane presents a major challenge in drug discovery, especially for biologics such as peptides, proteins, and nucleic acids. One potential strategy to subvert the membrane barrier and deliver biologics into cells is to attach them to "cell penetrating peptides" (CPPs). Despite three decades of investigation, the fundamental basis for CPP activity remains elusive. CPPs that enter cells via endocytosis generally exit from endocytic vesicles in order to reach the cytosol. Unfortunately, the endosomal membrane has proven to be a significant barrier towards cytoplasmic delivery of these CPPs such that often a negligible fraction of the peptides escapes into the cell interior. What are thus needed are new scaffolds and structures that impart peptides with highly proficient intrinsic cell penetrating ability to various cell types. Several naturally occurring polyketides and peptides exhibit remarkable cell permeability (e.g. cyclosporine and amanitins). These peptides are characterized by specific modifications (e.g., N-methylation of the backbone and cyclization) that can play a crucial role in their cell membrane permeability. The compositions and methods disclosed herein describe methods and approaches that enable the general utilization of similar modifications to generate compositions that may be of high therapeutic value and that may be capable of disrupting select protein-protein interactions with high selectivity.

Definitions

(22) As used herein, "reporter gene" refers to a gene whose expression can be assayed. Such genes include, for example, LacZ, β -glucuronidase (GUS), amino acid biosynthetic genes, the yeast LEU2, HIS3, LYS2, or URA3 genes, nucleic acid biosynthetic genes, the mammalian chloramphenicol transacetylase (CAT) gene, the green fluorescent protein (GFP) or any surface antigen gene for which specific antibodies are available. Reporter genes can result in both positive and negative selection.

(23) An "allele" refers to a DNA sequence of a gene which includes a naturally occurring, or pathogenic variant of a gene. Expression of differing alleles may lead to different protein variants.

(24) A "promoter" is a DNA sequence located proximal to the start of transcription at the 5' end of an operably linked transcribed sequence. The promoter can contain one or more regulatory elements or modules, which interact in modulating transcription of the operably linked gene.

Promoters can be switchable or constitutive. Switchable promoters allow for reversible induction or repression of operably linked target genes upon administration of an agent. Examples of switchable promoters include but are not limited to the LexA operator and the alcohol dehydrogenase I (alcA) gene promoter. Examples of constitutive promoters include the human beta-actin gene promoter.

(25) "Operably linked" describes two macromolecular elements arranged such that modulating the activity of the first element induces an effect on the second element. In this manner, modulation of the activity of a promoter element can be used to alter or regulate the expression of an operably-linked coding sequence. For example, the transcription of a coding sequence that is operably-linked to a promoter element can be induced by factors that activate the promoter's activity; transcription of a coding sequence that is operably-linked to a promoter element can be inhibited by factors that repress the promoter's activity. Thus, a promoter region is operably-linked to the coding sequence

of a protein if transcription of such coding sequence activity is influenced by the activity of the promoter.

(26) “In frame” as used herein throughout, refers to the proper positioning of a desired sequence of nucleotides within a DNA fragment or coding sequence operably linked to a promoter sequence, thereby permitting transcription and/or translation.

(27) “Fusion construct” refers to recombinant genes that encode fusion proteins.

(28) A “fusion protein” is a hybrid protein, i.e., a protein that has been constructed to contain domains from at least two different proteins. Fusion proteins described herein can be a hybrid proteins that possess both (1) a transcriptional regulatory domain from a transcriptional regulatory protein or a DNA binding domain from a DNA binding protein and (2) a heterologous protein to be assayed for interaction status. The protein that is the source of the transcriptional regulatory domain may different from the protein that is the source of the DNA binding domain. In other words, the two domains may be heterologous to each other.

(29) A transcriptional regulatory domain of a prey fusion protein can either activate or repress transcription of target genes, depending on the biological activity of the domain. Bait proteins of the disclosure may also be fusion proteins, where the fusion protein is encoded by a fusion gene that can encodes for a protein of interest that is operably linked to a DNA binding moiety.

(30) “Bridging interaction” refers to an interaction between a first protein and a second that occurs only when one or both of the first protein and the second protein interact with a molecule, such as a peptide or small molecule from a library. In some cases, the bridging interaction between the first protein and the second protein is direct, while in other cases the bridging interaction between the first protein and the second protein is indirect.

(31) “Expression” is the process by which the information encoded within a gene is revealed. If the gene encodes a protein, then expression involves both transcription of the DNA into mRNA, the processing of mRNA (if necessary) into a mature mRNA product, and translation of the mature mRNA into protein.

(32) As used herein, a “cloning vehicle” is any entity that is capable of delivering a nucleic acid sequence into a host cell for cloning purposes. Examples of cloning vehicles include plasmids or phage genomes. A plasmid that can replicate autonomously in the host cell is especially desired. Alternatively, a nucleic acid molecule that can insert (integrate) into the host cell's chromosomal DNA is useful, especially a molecule that inserts into the host cell's chromosomal DNA in a stable manner, that is, a manner that allows such molecule to be inherited by daughter cells.

(33) Cloning vehicles are often characterized by one or a small number of endonuclease recognition sites at which such DNA sequences may be cut in a determinable fashion without loss of an essential biological function of the vehicle, and into which DNA may be spliced in order to bring about its replication and cloning.

(34) The cloning vehicle can further contain a marker suitable for use in the identification of cells transformed with the cloning vehicle. For example, a marker gene can be a gene that confers resistance to a specific antibiotic on a host cell.

(35) The word “vector” can be used interchangeably with “cloning vehicle.”

(36) As used herein, an “expression vehicle” is a vehicle or vector similar to the cloning vehicle that is especially designed to provide an environment that allows the expression of the cloned gene after transformation into the host. One manner of providing such an environment is to include transcriptional and translational regulatory sequences on such expression vehicles, such transcriptional and translational regulatory sequences being capable of being operably linked to the cloned gene. Another manner of providing such an environment is to provide a cloning site or sites on such vehicle, wherein a desired cloned gene and a desired expression regulatory element can be cloned.

(37) In an expression vehicle, the gene to be cloned is usually operably-linked to certain control sequences such as promoter sequences. Expression control sequences will vary depending on

whether the vector is designed to express the operably-linked gene in a prokaryotic or eukaryotic host and can additionally contain transcriptional elements such as enhancer elements, termination sequences, tissue-specificity elements, or translational initiation and termination sites.

(38) A “host” refers to any organism that is the recipient of a cloning or expression vehicle. The host may be a yeast cell or a cultured animal cell such as a mammalian or insect cell. The yeast host may be *Saccharomyces cerevisiae*.

(39) A “host cell” as described herein can be a bacterial, fungal, or mammalian cell or from an insect or plant. Examples of bacterial host cells are *E. coli* and *B. subtilis*. Examples of fungal cells are *S. cerevisiae* and *S. pombe*. Non-limiting examples of mammalian cells are immortalized mammalian cell lines, such as HEK293, A549, HeLa, or CHO cells, or isolated patient primary tissue cells that have been genetically immortalized (such as by transfection with hTERT). Non-limiting example of the plant is *Nicotiana tabacum* or *Physcomitrella patens*. A non-limiting example of insect cell is a sf9 (*Spodoptera frugiperda*) cell.

(40) A “DNA-binding domain (DBD),” or a “DNA-binding moiety” is a moiety that is capable of directing specific polypeptide binding to a particular DNA sequence (i.e., a “protein binding site”). These proteins can be homodimers or monomers that bind DNA in a sequence specific manner. Exemplary DNA-binding domains of the disclosure include LexA, cI, glucocorticoid receptor binding domains, and the Ume6 domain.

(41) A “gene activating moiety” or “activation domain” (“AD”) is a moiety that is capable of inducing (albeit in many instances weakly inducing) the expression of a gene to whose control region it is bound (one example is an activation domain from a transcription factor). As used herein, “weakly” is meant below the level of activation effected by GAL4 activation region II and is preferably at or below the level of activation effected by the B42 activation domain. Levels of activation can be measured using any downstream reporter gene system and comparing, in parallel assays, the level of expression stimulated by the GAL4 region II-polypeptide with the level of expression stimulated by the polypeptide to be tested.

(42) Screening for Disruptors to a Protein-Protein Interaction (PPI)

(43) The often large and broad surfaces that can form the contact interface between two proteins can be potential targets of canonical small molecule inhibitors. However, the large and broad surfaces can have size limitations, and evolved resistance can occur readily. The specificity of antibodies can be combined with cell permeability in the form of short peptides, for example, peptides of less than 25 residues. Screening for short peptide disruptors of protein-protein interactions (PPIs) can be performed using technologies such as phage display or mRNA display. However, these screens are performed in vitro and require the purification of one of the interacting proteins of interest. Upon selection of a peptide sequence with affinity toward one of the proteins, secondary screens can be performed to validate that the peptide interferes with the binding interface of the second protein. This secondary screening can further rely upon the proper folding of the proteins and the replication of intracellular biophysical conditions in the assays.

(44) Methods and systems of the disclosure can involve the intracellular selection of peptide disruptors of PPIs. Stated differently, various systems described herein can be used to screen for molecules that selectively disrupt an interaction between two proteins. A model organism, for example *Saccharomyces cerevisiae*, can be employed, and the coexpression of a PPI of interest with a test DNA molecule comprising a DNA sequence that encodes a randomized peptide library can allow for the selection of unbiased peptides that interfere with a specific PPI using selection mechanisms (e.g., a stringent viability readout selection mechanisms). The method can involve a permutation of a yeast two-hybrid system that can rely on the reconstitution of a transcription factor that requires an interaction between one or two test proteins fused to one or two DNA binding domain(s) (DBDs) and a second test protein fused to a transcription activation domain (AD) or gene activating moiety.

(45) Methods and systems of the disclosure can use the reconstitution of a transcription factor

mediated by the interaction between a protein fused to an AD, for example, VP16, NF- κ B AD, VP64AD, BP64 AD, B42 acidic activation domain (B42AD), or p65 transactivation domain (p65AD) and another protein fused to a DBD, for example, LexA, cI, Gli-1, YY1, glucocorticoid receptor binding domain, or Ume6 domain.

(46) Methods and system of the disclosure can also use two different proteins, or two variants of one protein, fused to different DBDs. These proteins may interact with the same protein fused to an AD to drive two different or identical reporters. The system can identify inhibitors against a specific PPI in a complex without affecting the rest of the complex integrity (see FIGS. 1A and 1B). This system can also be used to identify selective inhibitors that disrupt a PPI between a specific isoform and its binding partner without affecting another variant (see FIGS. 2A and 2B).

(47) An efficient interaction between the two proteins of interest can direct RNA polymerase to a specific genomic site, and allow for the expression of a genetic element. The genetic element can be, for example, a gene that encodes a protein that enables an organism to grow on selection media. The selection media can be specific to, for example, ADE2, URA3, TRP1, KANR, or NATR, and will lack the essential component (Ade, Ura, Trp) or include a drug (G418, NAT). Markers that can detect when an interaction is no longer present (for example when the interaction is disrupted by an external composition) can be referred to as counter-selection markers, such as the URA3 gene, and can be poor or leaky (easily masked by the selection of mutants that escape the selection). This leakiness of the selection marker can lead to a high false positive rate.

(48) Methods and systems of the disclosure can combine a strong negative selection marker with the intracellular stabilization of the production of short peptides to screen for blockers of PPIs. An inducible two-hybrid approach can be employed, which can drive the expression of any one or combination of several cytotoxic reporters (death agents) as well as positive selection markers. A method of the disclosure involving induced expression of a combination of cytotoxic reporters in a two-hybrid system can allow for a multiplicative effect in lowering the false-positive rate of the two-hybrid assay, as all of the cytotoxic reporters must simultaneously be “leaky” to allow for an induced cell to survive.

(49) Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively disrupts an interaction between a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell; wherein a sequence of gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein, wherein a positive selection reporter is disposed within the host and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein, wherein, in the absence of the molecule, the interaction between the first test protein and the second test protein causes the gene activating moiety to activate expression of the death agent, while the interaction between the second test protein and the third test protein causes the gene activating moiety to activate the expression of the positive selection reporter.

(50) FIGS. 1A and 1B show a platform to identify a compound that disrupts a protein-protein interaction within a complex in a specific manner. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. A, B, and C refer to three proteins, wherein B and C each interact with A. Broken arrows indicate active expression of the reporter. +ve refers to positive selection markers, -ve refers to death agents (negative selection markers). Pep refers to a peptide, such as a peptide from a library. In some embodiments, a peptide library may be replaced with a library that includes compounds other than peptides like small molecules. In some embodiments, the small molecules are peptidomimetics. Two scenarios are shown; FIG. 1A illustrates a case where a peptide (e.g., a peptide from a library) is unable to disrupt the pairwise

interaction of interest (A-C), and a death agent is expressed, leading to cell death. FIG. 1B illustrates a case where a peptide (e.g., a peptide from a library) is able to disrupt the pairwise interaction of interest (A-C) without disrupting the A-B interaction. Selective peptide disruption activity is assayed by survival due to (1) the absence of expression of the death agent and (2) expression of the positive selection reporter (which provides evidence of selectivity).

(51) FIGS. 2A and 2B show a platform to identify a compound that disrupts a protein-protein interaction in a variant-specific manner. In other words, in contrast to FIGS. 1A and 1B in which B and C were unrelated protein, FIGS. 2A and 2B describe an analogous assay in which B and B' are related (but different) proteins (e.g., protein variants). As in FIGS. 1A and 1B, DBD1 and DBD2 are promoter-specific DNA-binding domains. AD refers to an activation domain. A, B, and B' refer to three proteins, wherein B and B' are two variants that are configured to interact with A. Broken arrows indicate active expression of the reporter. +ve refers to a positive selection marker, while -ve refers to a death agent. Pep refers to a peptide. Two scenarios are shown; FIG. 2A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest (A-B'), and a death is expressed, leading to cell death. FIG. 2B illustrates a case where a peptide is able to disrupt the pairwise interaction between A and B' without disrupting the interaction between the A and B. Selective peptides disruption is assayed by survival due to (1) the absence of expression of the death agent and (2) and expression of the positive selection reporter (which provides evidence of selectivity).

(52) Screening for Facilitators to Protein-Protein Interaction

(53) The system can additionally be used to screen for molecules that “bridge” an interaction between two proteins in a selective manner. In some embodiments, the system can be used to identify molecules which can bind to one isoform, or one protein, and bridge its interaction with another macromolecule, such as a protein, RNA, or DNA. For example, the bridging could occur to link the protein to an E3 ligase to mediate its degradation. For example, bridging can occur between an oncogenic protein such as K-Ras oncogenic alleles, Cyclin D family, Cyclin E family, c-MYC, EGFR, HER2, PDGFR, Raf kinase, VEGF and beta-catenin, or oncogenic variants such as IDH1(R132H, R132S, R132C, R132G, and R132L) or IDH2(R140Q, R172K), and an E3 ligase. An E3 ligase can be chosen from a list including, but not limited to, Cereblon, Skp2, MDM2, FBXW7, DCAF15, VHL, AMFR, ANAPC11, ANKIB1, ARL1, ARL11, ARL12, BARD1, BAF, BIRC2, BIRC3, BIRC7, BIRC8, BMI1, BRAP, BRCA1, CBL, CBLB, CBLC, CBLL1, CCDC36, CCNB1IP1, CGRRF1, CHFR, CNOT4, CUL9, CYHR1, DCST1, DTX1, DTX2, DTX3, DTX3L, DTX4, DZIP3, E4F1, FANCL, G2E3, HACE1, HECTD1, HECTD2, HECTD3, HECTD4, HECW1, HECW2, HERC1, HERC2, HERC3, HERC4, HERC5, HERC6, HLTF, HUWE1, IRF2BP1, IRF2BP2, IRF2BPL, Itch, KCMF1, KMT2C, KMT2D, LNX1, LNX2, LONRF1, LONRF2, LONRF3, LRSAM1, LTN1, MAEA, MAP3K1, MARCH1, MARCH10, MARCH11, MARCH2, MARCH3, MARCH4, MARCH5, MARCH6, MARCH7, MARCH8, MARCH9, MDM4, MECOM, MEX3A, MEX3B, MEX3C, MEX3D, MGRN1, MIB1, MIB2, MID1, MID2, MKRN1, MKRN2, MKRN3, MKRN4P, MNAT1, MSL2, MUL1, MYCBP2, MYLIP, NEDD4, NEDD4L, NEURL1, NEURL1B, NEURL3, NFX1, NFXL1, NHLRC1, NOSIP, NSMCE1, PARK2, PCGF1, PCGF2, PCGF3, PCGF5, PCGF6, PDZRN3, PDZRN4, PELI1, PELI2, PELI3, PEX10, PEX12, PEX2, PHF7, PHRF1, PJA1, PJA2, PLAG1, PLAGL1, PML, PPIL2, PRPF19, RAD18, RAG1, RAPSN, RBBP6, RBCK1, RBX1, RC3H1, RC3H2, RCHY1, RFFL, RFPL1, RFPL2, RFPL3, RFPL4A, RFPL4AL1, RFPL4B, RFWD2, RFWD3, RING1, RLF, RLIM, RMND5A, RMND5B, RNF10, RNF103, RNF11, RNF111, RNF112, RNF113A, RNF113B, RNF114, RNF115, RNF121, RNF122, RNF123, RNF125, RNF126, RNF128, RNF13, RNF130, RNF133, RNF135, RNF138, RNF139, RNF14, RNF141, RNF144A, RNF144B, RNF145, RNF146, RNF148, RNF149, RNF150, RNF151, RNF152, RNF157, RNF165, RNF166, RNF167, RNF168, RNF169, RNF17, RNF170, RNF175, RNF180, RNF181, RNF182, RNF183, RNF185, RNF186, RNF187, RNF19A, RNF19B, RNF2, RNF20, RNF207, RNF208, RNF212, RNF212B,

RNF213, RNF214, RNF215, RNF216, RNF217, RNF219, RNF220, RNF222, RNF223, RNF224, RNF225, RNF24, RNF25, RNF26, RNF31, RNF32, RNF34, RNF38, RNF39, RNF4, RNF40, RNF41, RNF43, RNF44, RNF5, RNF6, RNF7, RNF8, RNFT1, RNFT2, RSPRY1, SCAF11, SH3RF1, SH3RF2, SH3RF3, SHPRH, SIAH1, SIAH2, SIAH3, SMURF1, SMURF2, STUB1, SYVN1, TMEM129, TOPORS, TRAF2, TRAF3, TRAF4, TRAF5, TRAF6, TRAF7, TRAIIP, TRIM10 TRIM11 TRIM13 TRIM15 TRIM17 TRIM2 TRIM21 TRIM22 TRIM23 TRIM24 TRIM25 TRIM26, TRIM27, TRIM28, TRIM3, TRIM31, TRIM32, TRIM33, TRIM34, TRIM35, TRIM36, TRIM37 TRIM38 TRIM39 TRIM4 TRIM40 TRIM41 TRIM42 TRIM43 TRIM43B TRIM45 TRIM46, TRIM47, TRIM48, TRIM49, TRIM49B, TRIM49C, TRIM49D1, TRIM5, TRIM50, TRIM51, TRIM52, TRIM54, TRIM55, TRIM56, TRIM58, TRIM59, TRIM6, TRIM60, TRIM61, TRIM62, TRIM63, TRIM64, TRIM64B, TRIM64C, TRIM65, TRIM67, TRIM68, TRIM69, TRIM7, TRIM71, TRIM72, TRIM73, TRIM74, TRIM75P, TRIM77, TRIM8, TRIM9, TRIML1, TRIML2, TRIP12, TTC3, UBE3A, UBE3B, UBE3C, UBE3D, UBE4A, UBE4B, UBOX5, UBR1, UBR2, UBR3, UBR4, UBR5, UBR7, UHRF1, UHRF2, UNK, UNKL, VPS11, VPS18, VPS41, VPS8, WDR59, WDSUB1, WWP1, WWP2, XIAP, ZBTB12, ZFP91, ZFPL1, ZNF280A, ZNF341, ZNF511, ZNF521, ZNF598, ZNF645, ZNRF1, ZNRF2, ZNRF3, ZNRF4, Zswim2, and ZXDC. The peptide-mediated bridging event can be specific to a mutant variant, or to one member of a complex, without disrupting the integrity of the WT variant or the rest of the complex.

(54) Disclosed herein, in certain embodiments, is a method for identifying a molecule that selectively facilitates an interaction between a first test protein and a second test protein comprising: expressing in the host cell a first fusion protein comprising the first test protein and a DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising the third test protein and a different DNA-binding moiety; and delivering a molecule from a library to the host cell such that the molecule forms a bridging interaction between the first test protein and the second test protein; wherein a sequence of a gene for expressing a death agent is disposed within the host cell and operably linked a promoter DNA sequence specific for the DNA binding moiety of the third fusion protein; wherein a positive selection reporter is disposed within the host cell and operably linked to a promoter DNA sequence specific for the DNA binding moiety of the first fusion protein; and wherein the first test protein and second test protein to form a functional transcription factor that activates expression of the death agent when the molecule from the library forms the bridging interaction.

(55) FIGS. 3A and 3B show a platform to identify a compound that bridges a protein-protein interaction in a variant- or protein-specific manner. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. A, B, and B' refer to three proteins, wherein B and B' are two variants to be assayed for interaction with A. Broken arrows indicate active expression of the reporter. +ve refers to positive selection markers, -ve refers to death agents. Pep refers to a peptide (e.g., a peptide from a library). Two scenarios are shown; FIG. 3A illustrates a case where a peptide is able to bridge an interaction between one variant (B) and a protein (A). In this case, this is the control variant (B) and it leads to expression of the death and cell death. FIG. 3B illustrates a case where a peptide is able to bridge the secondary variant (B') and a protein (A). In this case, the peptide is bridging the variant of interest (B') and enables its binding to protein A. This combination of results leads to (1) the activation of a positive selection marker and (2) the lack of activation of the death agent (due to the lack of bridging to the unintended variant).

(56) In some embodiments, the host cell disclosed herein further comprises an integrated DNA encoding the first fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein; a plasmid DNA encoding the death agent; and a plasmid DNA encoding a positive selection reporter.

(57) To identify peptides that can disrupt or facilitate a PPI, a PPI integration plasmid (PLASMID

1; FIG. 8), a selection and library plasmid (PLASMID 2; FIG. 9), and a confirmation plasmid (PLASMID 3; FIG. 10) can be used. The integration of PLASMID 1 into the genome of the host cell (as confirmed using PLASMID 3) can be followed by transformation of a library of PLASMID 2 encoding random peptides with, for example, NNK or NNN codons.

(58) Expression of Fusion Proteins for PPI

(59) In some embodiments, the host cell disclosed herein comprises a plasmid vector, which comprises the components of PLASMID 1 (FIG. 8), or any combination of the components of PLASMID 1. PLASMID 1 can contain, for example, two restriction sites that enable the integration of two proteins that constitute the PPI of interest. The PPI of interest can involve a pair of domains having known importance for carcinogenesis, such as p53-MDM2, RAS-RASBDPs, and MYC-MAX. The PPI of interest can also involve the interaction of an oncogene (such as Cyclin E family, Cyclin D family, c-MYC, EGFR, HER2, K-Ras, PDGFR, Raf kinase, and VEGF) or a tumor suppressor (such as BRCA1, BRCA2, cyclin-dependent kinase inhibitor 1C, PTEN, p16, p27, p53, p73, and Retinoblastoma protein (pRb)) with a known cellular interaction partner. The PPI of interest can involve the interaction of a protein involved in the DNA repair pathway (such as ATM, ATRX, BRCA1, BRCA2, ERCC1, FANCB, FANCF, FEN1, HMGA1, HMGA1, MDC1, MGMT, MLH1, MSH2, MSH4, Mre11A, NBS1, NEIL1, PARP1, PARP2, PMS2, RAD51, RAD52, RAD54, RAD51AP1, WRN, and XPF) with another cellular factor.

(60) PLASMID 1 can be configured to express two proteins that constitute a PPI and an additional factor, for example, a variant of one of the proteins, such as KRAS (G12D, G12V, G12C, G12S, G13D, Q61K, or Q61L, etc.) and WT KRAS along with BRAF. The additional factor can also be another protein bound to one of the components of the PPI, or as member of a larger complex (such as YAP or TAZ disruption from TEAD without compromising VGLL4 binding to TEAD, or maintaining binding of BAX to BAK but preventing binding of BAX to BCL-2).

(61) In some embodiments, the host cell disclosed herein comprises PLASMID 1, wherein a DNA sequence encoding a first polypeptide is inserted in frame with Gal4-DBD, a second polypeptide is inserted in frame with LexA-DBD, and wherein a DNA sequence encoding a third polypeptide is inserted in frame with VP64-AD.

(62) In some embodiments, the first test protein is a variant of KRAS, the second test protein is c-Raf, and the third test protein is KRAS.

(63) In some embodiments, the first test protein is YAP or TAZ, the second test protein is TEAD, and the third test protein is VGLL4.

(64) PLASMID 1 can encode for the fusion of an activation domain or another gene activating moiety and a DBD to each protein driven by either a strong promoter and terminator (such as ADH1), or by an inducible promoter (such as GAL1). Other exemplary activation domains include those of VP16 and B42AD. In some embodiments, the DNA binding moiety is derived from LexA, TetR, LacI, Gli-1, YY1, glucocorticoid receptor, or Ume6 domain and the gene activating moiety is derived from Gal4, B42, or VP64, Gal4, NF- κ B AD, Dof1, BP64, B42, or p65. Each protein fusion can be tagged for subsequent biochemical experiments with, for example, a FLAG, HA, MYC, or His tag. PLASMID 1 can also include bacterial selection and propagation markers (i.e. ori and AmpR), and yeast replication and selection markers (i.e. TRP1 and CEN or 2 μ m). The plasmid may contain multiple bait proteins fused to different DBDs. The plasmid can also be integrated into the genome at a specified locus.

(65) Disclosed herein, in certain embodiments, is a library of plasmid vectors, each plasmid vector comprising: a DNA sequence encoding a different peptide sequence operably linked to a first switchable promoter; a DNA sequence encoding a death agent under control of a second switchable promoter; and a DNA sequence encoding a positive selection reporter under control of a third switchable promoter.

(66) Expression of Selection Markers

(67) Positive Selection Markers

(68) An efficient interaction between the two test proteins can direct RNA polymerase to a specific genomic site, and allow expression of a protein that enables an organism to grow on selection media. The selection media can be specific to, for example, ADE2, URA3, TRP1, KAN^{sup}.R, or NAT^{sup}.R, and can lack the essential component (Ade, Ura, Trp) or can include a drug (G418, NAT). PLASMID 2 (FIG. 9) can encode for one or more positive selection markers that enable an organism to grow on selection media.

(69) FIG. 6 shows the results of two systems for identifying a compound that specifically disrupts a protein-protein interaction in cell culture using a positive selection marker. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in yeast cells. In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintain interaction to drive expression of nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^{sup}.4 cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30° C. The results showed that the cells grown on media that selected for nutritional reporter 2 had particularly poor survival rate when the inhibitor was added, illustrating both (1) the specificity of the inhibitor for KRas(G12D) and c-Raf and (2) the validity of the screening assay.

(70) In the second platform, VGLL4 and YAP fused to DBDs and TEAD fused to AD were expressed in cells. In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintain interaction to drive expression of nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^{sup}.4 cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30° C. The results showed that the cells grown on media that selected for nutritional reporter 2 had particularly poor survival rate when the inhibitor was added, illustrating both (1) the specificity of the inhibitor for the YAP and TEAD interaction and (2) the validity of the screening assay.

(71) Negative Selection Markers

(72) An inducible two-hybrid approach can be employed, which can drive the expression of any one or combination of several cytotoxic reporters (death agents) as well as positive selection markers. A method of the disclosure involving induced expression of a combination of cytotoxic reporters in a two-hybrid system can allow for a multiplicative effect in lowering the false-positive rate of the two-hybrid assay, as all of the cytotoxic reporters must simultaneously be “leaky” to allow for an induced cell to survive. The cytotoxic reporters can be, for example:

(73) TABLE-US-00001 TABLE 1 Cholera toxin SEQ ID
MVKIIFVFFIFLSSFSYANDDKLYRADSRPPDEIKQSGGLMPRGQSEYFDRGTQMN (CtxA)
NO.: 1 INLYDHARGTQTGFVRHDDGYVSTSISLRSAHLVGQTILSGHSTYYIYVIATAPNM
FNVNDVLGAYSPHPDEQEVSALGGIPYSQIYGWYRVHFGVLDEQLHRNRGYRDRYY
SNLDIAPAADGYGLAGFPPEHRAWREEPWIIHAPPGCGNAPRSSMSNTCDEKTQSL
GVKFLDEYQSKVKRQIFSGYQSDIDTHNRIKDEL SpvB toxin SEQ ID
MLILNGFSSATLALITPPFLPKGGKALSQSGPDGLASITLPLPISAERGFAPALAL (*Salmonella*
NO.: 2

HYSSGGGNGPFGVGWSCATMSIARRTSHGVPQYNDSDFLGPDGEVLVQTLSTGDA
enterica)

PNPVTCTFAYGDVSFPQSYTVTRYQPRTESSFYRLEYWVGNSNGDDFWLLHDSNGIL
HLLGKTAAARLSDPQAASHTAQWLVEESVTPAGEHIYYSYLAENGDNVDLNGNEAG
RDRSAMRYLSKVQYGNATPAADLYLWTSATPAVQWLFTLVFDYGERGVDPQVPPAF
TAQNSWLARQDPFSLYNYGFEIRLHRLCRQVLMFHHPDELGEADTLVSRLLEID
ENPILTQLCAARTLAYEGDGYRRAPVNNMMPPPPPPPPMMGGNSSRPSKSKWAIVE
ESKQIQALRYSAQGYSVINKYLRGDDYPETQAKETLLSRDYLSTNEPSDEEFKNA
MSVYINDIAEGLSSLPETDHRVVYRGLKLDKPALSDVLKEYTTIGNIIDKAFMST
SPDKAWINDTILNIYLEKGHKGRILGDVAHFKEAEMLFPPNTKLKIESIVNCGSQ
DFASQLSKLRLSDDATADTNRIKRIINMRVLNS CARDS SEQ ID
MSENYFQGHMPNPVRFVYRVDLRSPEEIFEHGFSTLGDVRNFFEHILSTNFGRSY toxin

NO.: 3 FISTSETPTAAIRFFGSWLREYVPEHPRRAYLYEIRADQHFYNARATGENLLDLMR

(*Mycoplasma*

QRQVVFDSGDREMAQMGIRALRTSFAYQREWFTDGPIAAANVRS AWLVD AVPVEPG

pneumoniae)

HAHHPAGRVVETTRINEPEMHNP HYQELQTQANDQPWLPTPGIATPVHLSIPQAAS
VADVSEGTSASLSFACPDWSPPSSNGENPLDKCIAEKIDNYNLQSLPQYASSVKEL
EDTPVYLRGIKTQKTFMLQADPQNNNVFLVEVNP KQKSSFPQTIFFWDVYQRICKL
DLTGAQISLSLTAFTTQYAGQLKVHLSVSAVNAV NQKWKMT PQDIAITQFRVSSEL
LGQTENGLFWNTKSGGSQHDLYVCPLKNPPSDLEELQIIVDECTTHAQFVTMRAAS
TFFVDVQLGWYWRGYYYTPQLSGWSYQMKTPDGQIFYDLKTSKIFFVQDNQNVFFL
HNKLNKQTGYSDWVVEWLKHDMNEDKD ENFKWYFSRDDLTIPSVEGLNFRHIRCYA
DNQQLKVIISGSRWGGWYSTYDKVESNVEDKILVKDGFDRF SpyA Toxin SEQ ID
MLKKRYQLAMILLSCFSLVWQTEGLVELFVCEHYERAVCEGTPAYFTFSDQKGAE

(*Streptococcus* NO.: 4

TLIKKRWGKGLVYPRAEQEAMAAAYTCQQAGPINTSLDKAKGKLSQLTPELRDQVAQ
pyogenes)

LDAATHRLVIPWNIVVYRYVYETFLRDIGVSHADLTSYYRNHQFNPHILCKIKLGT
RYTKHSFMSTTALKNGAMTHRPVEVRICVKKGAKAA FVEPYSAVPSEVELLFPRGC
QLEVVGAYVSQDHKKLHIEAYFKGSL HopU1 SEQ ID

MNINRQLPVSGSERLLTPDVGVSRQACSERHYSTGQDRHDFYRFAARLHVDAQCFCG
(*Pseudomonas* NO.: 5

LSIDDLMDKFSDKH FRAEHPEYRDVYPEECSAIYMHTAQDYSSHLVRGEIGTPLYR
syringae)

EVNNYLRLQHENS GREAEIDNHDEKLSPHIKMLSSALNRLMDVAAFRGTVYRGIRG
DLDTIARLYHLFDTGGRYVEPAFMSTTRIKDSAQVFEPGTPNNIAFQISLKR GADI
SGSSQAPSEEEIMLPMMSEFVIEHASALSEGKHLFVLSQI Chelt toxin SEQ ID
MKTII SLIFIMFPLFVSAHNGNFYRADSRSPNEIKDLGGLYPRGYDFFERGT PMS NO.: 6

ISLYDHARGAPSGNTRYDDGFVSTTTDIDSAHEIGQNILSGYTEYYIYLIAPAPNL
LDVNAVLGRYSPHPQENEYSALGGIPWTQVIGWYVVNNGVLD RNIHRNRQFRADLF
NNLSPALPSESYQFAGFEPEHPAWRQEPWINFAPPGCGRNVRLTKHINQQDCSNSQ
EELVYKKLQDLRTQFKVDKKLKL VNKTSSNNIIFPNHDFIREWVDLDGNGDLSYCG
FTVDS DGSRK RIVCAHNNGNFTYSSINISLSDYGWPKGQRFIDANGDGLVDYCRVQ
YVWTHLYCSLSLPGQYFSLDKDAGYLDAGYNN SRAWAKVIGTNKY SFCRLTSNGYI
CTDIDSYSTAFKDDDQGWADSRYWMDIDGNGGDDY CRLVYNWTHLRCNLQGKDGLW
KRVESKYLDGGYPSLRFKIKMTSNKDNYCRIVRNHRVMECAYVSDNGEFHNYS LNM
PFSLYNKNDIQFIDIDGDNRDDICRYNSAPNTMECYLNQDKSFSQNKLVLYLSAKP

ISSLGSGSSKIIRTFNSEKNSSAYCYNAGYGTLRCDEFVIY Certhrax SEQ ID
MKEIIRNLVRLDVRSDVDENSKKTQELVEKLPHEVLELYKNVGGEIYITDKRLTQH toxin
NO.: 7 EELSDSSHKDMFIVSSEGKSFPLREHFVFAKGGKEPSLIHAEDYASHLSSVEVYY

ELGKA IIRDTFPLNQKELGNPKFINAINEVNQQKEGKGVNAKADEDGRDLLFGKEL
KKNLEHGQLVDLDLISGNLSEFQHVF AKSFALYYEPHYKEALKSYAPALFN YMLEL
DQMR FKEISDDVKEKNKNVLD FKWYTRKAESWGVQTFKNWKENLTISEKDIITGYT
GSKYDPINEYLRKYDGEIIPNIGGDL DKKSKKALEKIENQIKNLDAALQKSKITEN
LIVYRRVSELQFGKKYEDYNLRQNGIINEEKVMELESNFKGQTFIQHNYMSTSLVQ
DPHQSYSNDRYPILLEITIPEGVHGAYIADMSEYPGQYEMLINRGYTFKYDKFSIV
KPTREEDKGKEYLKVNL SIYLGNLNREK EFV toxin SEQ ID

MSQLNKWQKELQALQKANYQETDNQLFN VYRQSLIDIKKRLKVYTENAESLSFSTR
NO.: 8 LEVERLFSVADEINAILQLNSPKVEKTIKGYS AKQAEQGYGLWYTLEQSQNIALS
MPLINHDIYIMNLVNAPVAGKRLSKRLYKYRDELAQNV TNNIITGLFEGKSYAEIAR
WINEETEASYKQALRIARTEAGRTQSVTTQKGYEEAKELGINIKKKWLATIDKHTR

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EQVNAMKDYTSDAFAKMNKILRNEKYNPREKPYLVNIIQNLDDAISKFKLKHDIIT
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GAYIGTNSSMKKESEFLLNRNLKYTVEIVDNILEVTILG ExoT SEQ ID
MHIQSSQQNPSFVAELSQAVAGRLGQVEARQVATPREAQQLAQRQEAPKGEGLLSR NO.:
9 LGAALARPFVAIIEWLGLKLLGSRAHAATQAPLSRQDAPPAASLSAAEIKQMMLQKA
LPLTLGGLGKASELATLTAERLAKDHTRLASGDGALRSLATALVGIRDGSLIEASR
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QVGHDAGYLSRDPSPVARSFAGLGTITTLFGRSGIDVSEISIEGDEQEILYDKGT
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LRMRGLDLA CdtB SEQ ID
MKKIICLFLSFNLA FANLENFNVGTWNLQGSSAATESKWSVSVRQLVSGANPLDIL NO.:
10 MIQEAGTLPRTATPTGRHVQQGGTPIDEYEWNLTLSRPDRVFIYYSRVDVGANRV
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DAH FANMPQVNWMIAGDFNRDPSTITSTVDRELANRIRVVFPTSATQASGGTLDYA
ITGNSNRQQTYTPPLLAAILMLASLRSHIVSDHFVPVNFRRKF Diptheria SEQ ID
MSRKL FASILIGALLGIGAPPSAHAGADDVVDSSKSFVMENFSSYHGTPGYVDSI toxin
NO.: 11
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SCINLDWDVIRDKTKTKIESLKEHGP IKNKMSESPNKT VSEEKAKQYLEEFHQ TAL
EHP ELS ELKT V TGTNPVFAGANYAAWAVNVAQVIDSETADNLEKTTAALSILPGIG
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ExoU/VipB SEQ ID
MKLAEIMTKSRKLKRNLLEISKTEAGQYSVSAP EHKGLVLSSGGGAKGISYLGMIQA NO.:
12 LQERGKIKNLTHVSGASAGAMTASILAVGMDIKDIKKLIEGLDITKLLDNSGVGFR
ARGDRFRNILDVIYMMQMKKHLESVQQPIPEQQMNYGILKQKIALYEDKLSRAGI
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ARMDSLVEATIGGKWLHATSSVLNREKVYYNIDNMIYINTGEVTTTNTSPTPEQRA
RAVKNGYDQTMQLLD SHKQTFDHP LMAILYIGHDKLKDALIDEKSEKEIFEASAHA
QAILHLQE QIVKEMNDGDYSSVQNYLDQIEDILTVD AKMDDIQKEKAFALCIKQVN
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MNRVSGSSSATWQAVNDLVEQVSERTTLSTTGYQTAMGRLNKPEKSDADALMTMRR
NO.: 13
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EGFSNCDIQALEIGLHLRHKLGISDFTIYSNRKLSHNYVVIHPSNAFPKGAI VDSW
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14 PMRVMGLDHDTELFRTTDSRYIKNDKLAGNPQSMASILMHEELRPNRFASHTGAQP

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NO.: 15

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TGSTGYTSRSITKPFVEKGLKHFIAKATHGEKALTPKQCMETLDNLLRKSITLNSDS
QFAAGQALLVFRQVYAGEDAWGDAERVILKSHYNRGTVLQDEADKIELSRPFSEQD
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NSSGIRHSLREIARGCFVGAGFTVQDGDDFYKQICKNASKQFYNG VopF SEQ ID
MFKISVSQQANVMSTSDTAQRSSLKISIKSICNKSLSKKLHTLAEKCRRFSSQELKE NO.: 16
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NDSVFVTIESSSELIVKQDGSIPPAPPLPGNIPPAPPLPSAGNIPTAPGLPKQKAT
TESVAQTS DNRSKLMEEIRQGVKL RATPKSSSTEKSASDPH SKLMKELINHGAKLK
KVSTSDIPVPPPLPAAFASKPTDGRSALLSEIAGESKDRLRKAGSSETLNVSQPTV
AESSIPEAYDLLLSDEM FNLS PKLSETELNTLADSLADYLFKAADIDWMQVIAEQT
KGSTQATSLKSQLEQAPEYVKAFCDEILKFPDCYKSADV ASPE SPKAGPSSVIDVA
LKRLQAGRNLRFSTIDAKGTNELKKGEAILESAINAARSVMTAEQKSALLSSNVKS
ATFKVFSELPCMEGF AEQNGKAAFNALRLAFYSSIQSGDTAQQDIARFMKENLATG
FSGYSYLG LTSRVAQLEAQLAALTTK YopJ SEQ ID

MIGPISQINISGGLSEKETSSLISNEELKNIITQLETDISDGSWFHKNYSRMDVEV NO.: 17
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IDYKHINGKTS LILFEPANFN SMGPAMLAIRTKTAIERYQLPDCHF SMVEMDIQRS
SSECGIFS FALAKKLYIERDSLLKIHEDNIKGILSDGENPLPHDKLDPYLPVTFYK
HTQGKKRLNEYLNTPQG VGT VVNKKNETIVNREDNNKSIVDGKELSVSVHKKRIA
EYKTLLKV AvrPtoB SEQ ID

MAGINGAGPSGAYFVGHTDPEPASGGAHGSSSGASSSNSPRLPAPPDAPASQARDR NO.:
18 REMLLRARPLSRQ TREWVAQGMPPTAEAGVPIRPQESAEAAAPQARAEERHTPEAD
AAASHVRTEGG RTPQALAGTSPRHTGAVPHANRIVQQLVDAGADLAGINTMIDNAM
RRHAIALPSRTVQSILIEHFP HLLAGELISGSELATAFRAALRREVRQQEASAPPR
TAARSSVRTPERSTVPPTSTESSSGSNQRTLLGRFAGL MTPNQRRPSSASNASASQ
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HRRPIPMDIAYALQGVGIAPSIDTGESLMENPLMNLSVALHREALGPRPARAQAPRP
AVPVAPATVSRRPDSARATRLQVIPAREDYENNVAYGVRLLSLNP GAGVRETVAAF
VNNRYERQAVVADIRAALNLSKQFNKLRTVSKADAASNKPGFKDLADHPDDATQCL
FGEELSLTSSVQQVIGLAGKATDMSESYSREANKDLVFMDMKKLAQFLAGKPEHPM
TRETNAENIAKYAFRIVP SdbA SEQ ID

MHKKYNYYSLEKEKKT FWQHILDILKAPFRLPGWVVSFFLARNITHVALNPNNIPQ NO.:
19 QRLIH LTKTSNRPEDDIVVINFKKRPPHKWENDTLIKIANTIAALPFVTPRLRTRL
HYDNENDINHVNKLLAEIDALVQGKSKQKYCKGRAFDWSKIHLKGLEFLDPKMRGY
VYEQLHEKYGYVSYTTKRKPNIEFFTLKTPDGSELDSVQVTGEDEEKKPMGERKFI
ITCIARDQNFINWIKDLNYTAKNLGATAISFN YRGVDYSRGLVWTENNLVDDILAQ
VQRLISLGADPKNICLDGMCIGGAVATIAAAKLHEKGMKVKLNNERSFTSLSSLVF
GFIVPELQTANWWSPLTYGRFLLAGVVYALLTPLIWL AGWPVDVTKAWNRIPAQDK
MYSVVRDKDNGLYDGV IHDHFCSIASLVDSQINSILYKLSTDQPLTEEEKQILCDD
QFSHHFKPSQSVLKNPKYKGPHFISRQDLVAELGHREEYTNHDYFLDRLREKFQLD
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DNVVITQHHAELYKNKPFSITSVLIRIGVWFTSLPILEDILKGVMRFIGYPVLPSS

SIFWDQMSKIQQSETKKENGIETGRTRPYVDMLLDIYPEGYEYTAFNNAATHLTSSI
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VKYYNTVFLPVYNAERGTSYQPIAIDQYMTDLPSLGCIFMNNLEELTSEQRQLME
IHAVNMSEPFKEAHFGKEQGFKAVHNIDPRNNPMIRNAFKDPSLTKEYLDKTQSFDL
HFNVYKKEKQNALPVLNGKEKITIKPHAKIASIMIGSLAANASADYAKYLLNQGYE
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NO.: 20

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NETLRAQIKQCFTQFESLKPRVAELGLTRASGAGTEVEAHESTVKEQEIISQNTVG
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LNTPVCGIPFSQWGTIGGAASAYVASGV DLTQAANEIKGLAQQMQKLLSLM SptP SEQ ID
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NO.: 25
PLFKNTEVVQKHTENIRVQDQKILQTF LHALTEKYGETAVNDALLMSRINMNKPLT
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QQTDIPTTHFHRGNASEGRAVLTSKTVKDFMLQKLNSLDIKGNASKDPAYARQTCE
AILS AVYSNNKDQCKLLISKGV SITPFLKEIGEAAQNAGLPGEIKNGVFTPGGAG
ANPFVVPLIASASIKYPHMFINH NQQVSFKAYA EKIVMKEVTPLFNKGTMTPTQQF
QLTIENIAN KYLQNAS SopB/SigD SEQ ID
MQIQSFYHSASLKTQEAFKSLQKTLYNGM QILSGQGKAPAKAPDARPEIIVLREPG NO.:
27 ATWGNYLQH QKASNHSLHNLYNLQRDLLTVAATVLGKQDPVLTSMANQMELAKVKA
DRPATKQEEAAAKALKKNLIELIAARTQQQDGLPAKEAHRFAAVAFRDAQVKQLNN
QPWQTIKNTLTHNGHHYTNTQLPAAEMKIGAKDIFPSAYEGKGVCSWDTKNIHHAN
NLWMSTVSVHEDGKDKTLFCGIRHGVLS PYHEKDLLRHVGAENKAKEVLTAA LFS
KPELLNKALAGEAVSLKLVS VGLLTASNIFGKEGTMVEDQMRAWQSLTQPGKMIHL
KIRNKDGD LQTVKIKPDVAAFNVGVNELALKLGFGLKASDSYNAEALHQLLGNDLR
PEARPGGWVGEWLAQYPDNYEVVNTLARQIKDIWKNNQH HKDGGEPYKLAQRLAML
AHEIDAVPAWNCKSGKDRTGMM DSEIKREIISLHQTHMLSAPGSLPDSSGGQKIFQK
VLLNSGNLEIQKQNTGGAGNKVMKNLSPEVLNLSYQKRVGDENIWQSVKGISSLIT S SipA
SEQ ID MVTSVRTQPPVIMPGMQTEIKTQATNLAANLSAVRESATTTLSGEIKGPQLED FPA
NO.: 28
LIKQASLDALFKCGKDAEALKEVFTNSNNVAGKKAIMEFAGLFRSALNATSDSPEA
KTLLMKVGAEYTAQIIKDGLKEKSAFGPWL PETKKA EAKLENLEKQLLDIIKNNTG
GELSKLSTNLVMQEVMPIYASCI EHNFGCTLDPLTRSNLTHLVDKAAAKAVEALDM
CHQKLTQE QGTSVGREARHLEMQTLIPLLLRN VFAQIPADKL PDPK IPEPAAGPVP
DGGKKA EPTGINININIDSSNHSVDNSKHINNSRSHVDNSQRHIDNSNHDNSR KTI
DNSRTFIDNSQRNGESHSTNSSNVSHSHSRVDSTTHQTETAHSASTGAIDHGIAG
KIDVTAHATAEAVTNASSESKDGKVVTSEK GTTGETTSFDEVDGVT SKSIIGKPVQ
ATVHGVDDNKQQSQTA EIVNVKPLASQLAGVENVKTDTLQSDTTVITGNKAGTTDN
DNSQTDK TGPFSGLKFKQNSFLSTVPSVTNMHSMHFDARETFLGVIRKALEPDTST
PFPVRRAFDGLRAEILPNDTIKSAALKAQCSDIDKHPELKAKMETLKEVITHHPQK
EKLA EIALQFAREAGLTRLKGETDYVLSNVLDGLIGDGSWRAGPAYESYLNKPGVD

RVITTVDTGLHMQR YpkA SEQ ID
MKSVMKIMGTMPPSISLAKAHERISQHWQNPVGELNIGGKRYRIIDNQVLRLNPHSG NO.:
29 FSLFREGVVGKIFSGKMFNFSIARNLTDTLHAAQKTTSQELRSDIPNALSNLFGAKP
QTELPLGWKGEP LSGAPDLEGMRVAETDKFAEGESHISIIETKDKQRLVAKIERSI
AEGHLFAELEAYKHIYKTAGKHPNLANVHGMAVVPYGNRKEEALLMDEV D GWRCSD
TLRTLADSWKQGKINSEAYWGTIKFIAHRLLDVTNHLAKAGVVHNDIKPGNVV FDR
ASGEPVVIDLGLHSRSGEQPKGFTE S FKAPELGVGNLGASEKSDVFLVVSTLLHCI
EGFEKNPEIKPNQGLRFITSEPAHVMDENGYP IHRPGIAGVETAYTRFITDILGVS
ADSRPDSNEARLHEFLSDGTIDEESA KQILKDTLTGEMSP LSTDVRRITPKKLREL
SDLLRTHLSSAATKQLDMGGVLSDLDTMLVALDKAEREGGV D K D Q L K S F N S L I L K T
YRVIEDYVKGREGDTKNSSTEVS PYHRSNFMLSIVEPSLQRIQKHL D QTHSFSDIG
SLVRAHKHLETLL E V L V T L S Q Q G Q P V S S E T Y G F L N R L A E A K I T L S Q Q L N T L Q Q Q Q E
SAKAQLSILINRSGSWADVARQSLQRFDSTRPVVKFGTEQYTAIHRQMMAAAHA AIT
LQEVSEFTDDMRNFTVDSIPLLIQLGRSSLMDEHLVEQREKLRELTTIAERLNRLE REWM
YopM SEQ ID
MFINPRNVSN TFLQEPLRHSSNLTEMPVEAENVKSKTEYYNAWSEWERNAPPGNGE
NO.: 30 QREMAVSRLRDCLDRQAHELELNNLGLSSLPELPPHLESLVASCNSLT ELP ELPQS
LKS LQVENNNLKALPDLPPSLKKLHVRENDLTDLPELPQSLES LRV D N N N L K A L S D
LPPSLEYLTASSNKLEELPELQNL PFLAAIYADNNLLETLPDLPPSLKKLHVREND
LTDLPELPQSLES LQVDNNNLKALSDLP SLEYLTASSNKLEELPELQNL PFLAAI
YADNNLLETLPDLPPHLEILVASYN S L T E L P E L P Q S L K S L R V D N N N L K A L S D L P P S
LEYLTASSNKLEELPELQNL PFLAAIYADNNLLETLPDLPPSLKKLHVRENDLTDL
PELPQSLTFLDVSDNNISGLSELPPNLYYLDASSNEIRSLCDLP PSLVDLNVKSNQ
LSELPALPPHLERLIASFNYLAEVPELPQNLKQLHVEQNALREFPDIPESLEELEM
DSERVVDPYEFAHETTDKLEDDVFE Amatoxin SEQ ID
MSDINATRLPIWGIGCNPCVGDDVTLLTRGEALC NO.: 31 Phallacidin SEQ ID
MSDINATRLPAWLVD C P C V G D D V N R L L T R G E S L C NO.: 32 Killer toxin SEQ ID
MIKPERSILTILIGILCLLAYVLANGEPHDGDNEWSSYCS D Q G F R R S D D G L V T T P D K P 1
NO.: 33
VGQESIGKNSINGSELVDYLQCLKVRLNGQKQVVSNDGWLLLLVQEPSVNV T Q K A M
SECNYNVSSGHKAGSYIQVTNTPADYKVISRRGSYEGDQLPEDVKPYFGVQKTS D Y
RPISKRINPNLT LRQLAYNF A A L N M C S L W C N S C I S R S C P Y Y I A E L T V H V N N I H G T
VWLHHFCRNAS P Q G G N L Y S T L T I S H K D T A Y Y V G T G W W K V R S T A A T T N D V A G D W Y P A
SWNQYWCGPHY Killer toxin SEQ ID
MLIFSVLMYLGLLLAGASALPNGLSPRNNAF C A G F G L S C K W E C W C T A H G T G N E L R Y K P 6
NO.: 34
ATAAGCGDHLSKSYDARAGHCLFSDDL RNQFYSHCSSLNNNMSCRSLSKRTIQDS
ATDTVDLGAELHRDDPPPTASDIGKRGRPRPVMCQCVDTTNGGVRLDAVTRAACS
IDSFIDGYYTEKDGF C R A K Y S W D L F T S G Q F Y Q A C L R Y S H A G T N C Q P D P Q Y E Killer Toxin
SEQ ID
MTKPTQVLVRSVSILFFITLLHLVVALNDVAGPAETAPVSLLPREAPWYDKIWEVK K1
NO.: 35
DWLLQRATDGNW G K S I T W G S F V A S D A G V V I F G I N V C K N C V G E R K D D I S T D C G K Q T L
ALLVSIFVAVTSGHHLIWGGNRPV S Q S D P N G A T V A R R D I S T V A D G D I P L D F S A L N D
ILNEHGISILPANASQYVKRSDTA E H T T S F V V T N N Y T S L H T D L I H H G N G T Y T T F T T
PHIPAVAKRYVYPMCEHG I K A S Y C M A L N D A M V S A N G N L Y G L A E K L F S E D E G Q W E T N
YYKLYWSTGQWIMSMKFIEESIDNANND F E G C D T G H Killer Toxin SEQ ID
MGHLAILFSIIAVLNIATAVASSDSIYLKGHRVGQDIDSLYRVYDNGTMYPVTFNE K28
(KHR) NO.: 36
WLNDLTGMNDLATNNATILKR D S S D V S C V T E T C Q Y V D Y H V D D E G V I T I D I S T Y R I P

VIEWDSGSAGASYGVSKRDTKYETFCFKKICGINVSGFCNAYDFAVHAFDEGGSVY
NPVSGITDRIKEATKRDKTECLGYELDHVRIDPAVDWSISISTWKQGSANCDTQAS
ADSLKCAAQKALESEHNHQKTAFCIHLDNNGGSFNLDIRLISELSFSKYNPWALPCP
KYKGSNSWQVVSDCFQ Killer Toxin SEQ ID
MPRFAIIFALLIAYSLFLSTLFTGSIPDRANTVTSNAPCQVVIWDWIRTRRICNCC K28
(KHS) NO.: 37

SRLCYSLGRSNLSRTAKRGVCTIAGAVLATAAVIVA AVLVGKSSGSATKRGLTKT
ISVLNHTIPFTDHILNGQTL SNGTGSNFVTIGFSGYAVHATIKRASTTDIISWVIP
ESMEPTLARVASYVSSSSINLA AVPDTGGNASALSFQNAVQEFATSWVSM TYDQSY
GDLRNVANDEGGEEILILMRKRSYRISFQVIETGSTALLLRTRRVVSQ LITMTYL
TVQARVGIQIGDIFQHYGGIDNYVMTSISVLR TLEDKAFHENKLLIVREPPNKS
DANQSYRLRPFSANDLIQNLKSVDIGFLAFC SFFDKYAHYPEIIMMKITIFISKGN
LWSIIVVIQARYVRKRVMKVRGQMPGGLLTNMESLLNIVSTPNLNISEFHIQTHSM
SQSKPMYFQKQCYSSQNNIIYIYNSIHITCGAVYVIVHDVRTPSV FVLIELRNCKP
LKNSWCETTKTSPRDTKIKKNEYNETVCRRAGALLDGRVRTIRFLMMRTHWSRVKG
VSCNTANRLSRFCNHVVSYYPSQNATIHLLPTSLRAESLEQQYTTRPLSSSNRFC
CLKSIFINNCKKACESPSLVSCNLQQTAE LLMVYYLYICEACYVSRNHDLLSKQCM
STVRAVYVARMRLPKERSTFPCMPRLCWL VNGVVVV Anthrax SEQ ID
MHVKEKEKNKDENKRKDEERNKTQEEHLKEIMKHIVKIEVKGEEAVKKEAAEKLLE
lethal factor NO.: 38

KVPSDVLEMYKAIGGKIYIVDGDITKHISLEALSEDKKKIKDIYGKDALLHEHYVY
endopeptidase
AKEGYEPVLVIQSSSEDYVENTEKALNVYYEIGKILSRDILSKINQPYQKFLDVLNT
IKNASDSDGQDLLFTNQLKEHPTDFSVEFLEQNSNEVQE VFAKAFAYYIEPQHRDV
LQLYAPEAFNYMDKFNEQEINLSLEELKDQRM LSRYEKWEKIKQHYQHWSDSLSEE
GRGLLKKLQIPIEPKKDDIIHSLSQEEKELLKRIQIDSSDELSTEEKEFLKKLQID
IRDSLSEEEKELLNRIQVDSSNPLSEKEKEFLKKLKL DIQPYDINQRLQDTGGLID
SPSINLDVRKQYKRDIQNIDALLHQSIGSTLYNKIYLYENMNINNL TATLGADLVD
STDNTKINRGIFNEFKKNFKYSSSNYMIVDINERPALDNERLKWRIQLSPDTRAG
YLENGKLILQRNIGLEIKDVQIIKQSEKEYIRIDAKVVPKSKIDTKIQEAQLNINQ
EWNKALGLPKYTKLITFNVHNRYASNIVESAYLILNEWKNNIQSDLIKKVTNYLVD
GNGRFVFTDITLPNIAEQYTHQDEIYEQVHSGLYVPESRSILLHGPSKGVELRND
SEGFIHEFGHAVDDYAGYLLDKNQSDLV TNSKKFIDIFKEEGSNLTSYGRTNEAEF
FAEAFRLMHSTDHAERLKVQKNAPKTFQFINDQIKFIINS Shiga Toxin SEQ ID
MKCILLKWVLCLLGFSSVSYSREFTIDFSTQQSYVSSLNSIRTEISTPLEHISQG NO.: 39
TTSVSVINHTPPGSYFAVDIRGLDVYQARFDHLRLIIEQNNLYVAGFVNTATNTFY
RFSDFAHISVPGVTTVSMTTDSSYTTLQRVAALERSGMQISRHS LVSSYLALMEFS
GNTMTRDASRAVLR FVIVTAEALRFRQIQREFRQALSETAPVYTMTPGDVDLTLNW
GRISNVLP EYRGEDGVRVGRISFNNISAILGT VAVILNCHHQGARSVRVAVNEESQP
ECQITGDRPVIKINNTLWESNTAA AFLNRKSQSLYTTGE Saporin SEQ ID
MKSWIMLVVTWLII LQTTVTAVIIEYELNLQGT TKAQYSTFLKQLRDDIKDPNLHYG Toxin
NO.: 40

GTNLPVIKRPVGP PKFLRVNLKASTGTVSLAVQRSNLYVAAYLAKNNNKQFRAYYF
KGFQITTNQLNNLFPEATGVSNNQQLGYGESYPQIQNAAGVTRQQAGLGIKKLAES
MTKVNGVARVEKDEALFLLIVVQMVGEAARFKYIENLV LNNFDTAKEVEPV PDRVI
ILENNWGLLSRAAKTANNGVFQTPLVLTSYAVPGVEWRVTTVAEVEIGIFLNVDNN
GLPSIYNNIISGAFGDTY Ricin Toxin SEQ ID
MYAVATWLCFGSTSGWSFTLEDNNIFPKQYPIINF TTAGATVQSYTNFIRAVRGRL NO.: 41
TTGADVRHDIPVLPNRVGLPINQRFILVELSNHAE LSVTLALDVTNAYVVG YRAGN
SAYFFHPDNQEDAEATHLFTDVQNRYTFAFGGNYDRLEQLAGNLRENIELGNGPL

EEAISLYSTYTLQPLTLARSIQMISEAAARFQYIEGEMRTRIIRYNRRSAP
DPSVITLENSWGRLSTAIQESNQGAFASPIQLQRRNGSKFSVYDVSILIPIIALMV
YRCAPPPSSQFSLIRPVVPNFNADVCMDEPIVRIVGRNGLCVDVRDGRFHNGNA
IQLWPCKSNTDANQLWTLKRDNTIRSNGKCLTTYGYSPGVYVMIYDCNTAATDATR
WQIWDNGTIINPRSSLVLAATSGNSGTTTLTVQTNIYAVSQGWLPTNNTQPFVTTIV
GLYGLCLQANSQGQVWIEDCSSEKAEQQWALYADGSIRPQQNRDNCLTSDSNIRETV
VKILSCGPASSGQRWMFKNDGTILNLYSGLVLDVRRSDPSLKQIILYPLHGDPNQI WLPLF

(74) In some embodiments, the death agent is an overexpressed product of genetic element selected from DNA or RNA. In some embodiments, the genetic element is a Growth Inhibitory (GIN) sequence such as GIN11.

(75) In some embodiments, the death agent is a ribosomally encoded xenobiotic agent, a ribosomally encoded poison, a ribosomally encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, or any combination thereof. In some embodiments, the ribosomally encoded death agent is Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phallacidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, or any combination thereof.

(76) Along with one or more positive selection markers, a plasmid (such as PLASMID 2) can also include one or more negative selection markers under control of a different DNA binding sequence to enable binary selection. The plasmid (e.g., PLASMID 2) can encode for one or more of negative selection markers in Table 1 driven by a promoter which depends on the DBD present in the PPI integration plasmid—DNA Binding Sequence (DBS), for example, the LexAop sequence (DBS) which can become bound by LexA (DBD). In some embodiments, to ensure repression of the ‘death agents,’ the plasmid (e.g., PLASMID 2) can include a silencing construct such as a TetR'-Tup11 fusion driven by a strong promoter (such as ADH1) to bind the DBD and silence transcription in the presence of doxycycline. The plasmid (e.g., PLASMID 2) can comprise bacterial selection and propagation markers (i.e. ori and AmpR), and yeast replication and selection markers (i.e. LEU2 and CEN or 2 um) as well.

(77) FIG. 7 shows two platforms to identify a disrupting compound in cell culture using both a negative and a positive selection marker. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in yeast cells. In the absence of inhibitors, the KRas(G12D) and c-Raf maintain an interaction to drive expression of a cytotoxic reporter, resulting in a low amount of cell growth/survival. A nutritional reporter was controlled by KRas and c-Raf interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30° C. In cell populations with KRas(G12D) and c-Raf interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the inhibitor for the KRas(G12D) and c-Raf interaction.

(78) In the second platform, VGLL4 and YAP fused to DBDs and TEAD fused to AD were expressed in cells. In the absence of inhibitors, the YAP and TEAD maintain an interaction to drive expression of cytotoxic reporter. A nutritional reporter was controlled by VGLL4 and TEAD interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30° C. In cell populations with YAP and TEAD interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the disruptor to YAP and TEAD interaction.

(79) In some embodiments, the host cell can further comprises more than one sequence for expressing a positive control reporter that is activated by a promoter DNA sequence specific for a

DNA binding moiety. In some embodiments, the host cell further comprises more than one sequence for expressing a death agent that is activated by a promoter DNA sequence specific for a DNA binding moiety.

(80) FIGS. 4A and 4B illustrate a platform with more than one positive selection markers and a negative selection marker for identifying a compound that disrupts a protein-protein interaction within a complex in a specific manner. The multiple selection markers can reduce false positive rate due to mutations that leads to avoidance of selection. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. Prey, BaitWT, and BaitMut refer to three proteins, wherein BaitWT and BaitMut each interacts with Prey. Broken arrows indicate active expression of the reporter. Markers 1 and 2 refer to positive selection markers, toxins refers to death agent. The ring comprising smaller circular items illustrates a potential inhibitor (e.g., a randomly produced peptide). Two scenarios are shown; FIG. 4A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest, and death agent is expressed, leading to cell death. FIG. 4B illustrates a case where a peptide is able to disrupt the Prey and BaitMut interaction by acting on the Prey without disrupting the BaitWT and Prey interaction. Peptide disruption activity is assayed by survival, and specificity is assayed by growth on the expressed positive selection reporter, marker 2.

(81) FIGS. 5A and 5B show a platform that is analogous to the platform of FIGS. 4A and 4B, where the inhibitor binds to the bait instead of the prey. More specifically, the platform of FIGS. 5A and 5B has more than one positive selection marker and one negative selection marker for identifying a compound that disrupts a protein-protein interaction within a complex in a specific manner. The multiple selection markers can reduce false positive rate due to mutations that leads to avoidance of selection. DBD1 and DBD2 are promoter specific DNA-binding domains. AD refers to an activation domain. Prey, BaitWT, and BaitMut refer to three proteins, wherein BaitWT and BaitMut each interacts with Prey. Broken arrows indicate active expression of the reporter. Markers 1 and 2 refers to positive selection markers, toxins refers to death agent. The ring comprising smaller circular items illustrates a randomly produced peptide. Two scenarios are shown; FIG. 5A illustrates a case where a peptide is unable to disrupt the pairwise interaction of interest, and death agent is expressed, leading to cell death. FIG. 5B illustrates a case where a peptide is able to disrupt the Prey and BaitMut interaction by acting on the BaitMut without disrupting the BaitWT and Prey interaction, and peptide disruption activity is assayed by survival and specificity is assayed by growth on the expressed positive selection reporter, marker 2.

(82) A plasmid (e.g., PLASMID 3) can be used to confirm expression of the reporters and the successful construction of the strains. PLASMID 3 can include a direct fusion between the AD and one or multiple DBDs. The plasmid (e.g., PLASMID 3) can further include bacterial selection and propagation markers (i.e. ori and AmpR), and yeast replication and selection markers (i.e. TRP1 and CEN or 2 um).

(83) FIG. 10 illustrates a confirmation plasmid. The depicted plasmid show the integration of two bait-prey fusion proteins, each with its own DBD. Protein tags may be included to enable detection of the proteins. The plasmid may also include propagation and selection markers for growth in hosts.

(84) Disclosed herein, in certain embodiments, is a library of plasmid vectors, each plasmid vector comprising a DNA sequence encoding a different peptide sequence operably linked to a first switchable promoter; a DNA sequence encoding a death agent under control of a second switchable promoter; and a DNA sequence encoding a positive selection reporter under control of a third switchable promoter.

(85) Addition or Expression of Modulators

(86) A molecule from a library that can selectively disrupt or facilitate PPI of interest can be screened by via use of positive and/or negative selection markers in a host cell.

(87) In some embodiments, the molecule is small molecule. In some embodiments, the small

molecule is peptidomimetic. The host cell can be made to become permeabilized to small molecules, for example by deletion of drug efflux pumps, such as PDR5, ERG6, or 12gene Δ 0HSR (Chinen, 2011), to enable a small molecule screening approach. The host cell can additionally carry mutations to enable more efficient transformation with vectors and/or more efficient uptake small molecules.

(88) In other embodiments, the molecule is peptide or protein. In some embodiments, the peptide or protein is derived from naturally occurring protein product. In another embodiment, the peptide or protein is synthesized protein product. In other embodiments, the peptide or protein is a product of recombinant genes.

(89) In some embodiments, the molecule is introduced to the host cell exogenously. In other embodiments, the molecule is the expression product of test DNA inserted into the host cell, wherein the test DNA comprises of DNA sequences that encodes a polypeptide. Libraries can be formed by delivery of a plurality of test DNA molecules into host cells. In some embodiments, the peptide sequences of the polypeptides in the library are random. In some embodiments, the different peptide sequences are pre-enriched for binding to a target.

(90) To screen for peptides that selectively disrupt or facilitate a PPI of interest, peptides from a randomized peptide library can be applied to the host cell. PLASMID 2 can be further used to express a randomized peptide library (such as a randomized NNK 60-mer sequences). PLASMID 2 can include a restriction site for integration of a randomized peptide library driven by a strong promoter (such as the ADH1 promoter) or an inducible promoter (such as the GAL1 promoter).

(91) In some embodiments, the randomized peptide library is about 60-mer. In some embodiments, the randomized peptide library is from about 5-mer to 20-mer. In some embodiments, the randomized peptide library is less than 15-mer.

(92) The library can also initiate with a fixed sequence of, for example, Methionine-Valine-Asparagine (MVN) for N-terminal stabilization and/or another combination of high-half-life N-end residues (see, for e.g., Varshavsky. Proc. Natl. Acad. Sci. USA. 93:12142-12149 (1996)) to maximize the half-life of the peptide, and terminate with the 3'UTR of a short protein (such as sORF1). The peptide can also be tagged with a protein tag such as Myc. In some embodiments, N-terminal residues of the peptide comprise Met, Gly, Ala, Ser, Thr, Val, or Pro or any combination thereof to minimize proteolysis.

(93) The plurality of different short peptide sequences can be randomly generated by any method (e.g. NNK or NNN nucleotide randomization). The plurality of different short peptide sequences can also be preselected, either by previous experiments selecting for binding to a target, or from existing data sets in the scientific literature that have reported rationally-designed peptide libraries.

(94) In some embodiments, the library comprises polypeptides about 60 amino acids or fewer in length. In another embodiment, the library comprises polypeptides about 30 or fewer amino acids in length. In another embodiment, the library comprises polypeptides about 20 or fewer amino acids in length.

(95) Modification of Disrupting or Facilitating Peptides

(96) The peptide that disrupts or facilitates PPI can also be a product of post-translational modification. The post-translational modification can include any one or combination of cleavage, cyclization, bi-cyclization, methylation, halogenation, glycosylation, acylation, phosphorylation, and acetylation. In some embodiments, the methylation comprises reacting with an N-methyltransferase. In some embodiments, the post-translational modification is done by naturally occurring enzymes. In some embodiments, the post-translational modification is done by synthetic enzymes. In some embodiments, the synthetic enzymes are chimeric.

(97) The peptide can be ribosomally synthesized and post-translationally modified peptide (RiPP) whereby the core peptide is flanked by prepeptide sequence comprising a leader peptide and recognition sequences which signal for the recruitment of maturation, cleavage, and/or modifying enzymes such as excision or cyclization enzymes including, for example, lanthipeptides maturation

enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP) patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and POPB from *Galerina marginata*, *Lentinula edodes*, *Omphalotaceae olearis*, *Dendrothele bispora*, or *Amanita bisporigera*, or other species. In some embodiments, the cyclization or bicyclization enzymes are synthetic chimeras.

(98) In one example, as illustrated in FIG. 11, the variable peptide library region is embedded within the primary sequence of a modifying enzyme (e.g., the homolog of the omphalotin N-methyltransferase enzyme from *Dendrothele bispora*, *Marasmius fiardii*, *Lentinula edodes*, *Fomitiporia mediterranea*, *Omphalotus olearius* or other) and contains random residues, some of which may be post-translationally decorated by additional modifications like hydroxylation, halogenation, glycosylation, acylation, phosphorylation, methylation, acetylation. This diversified variable region is excised and modified to form N-to-C cyclized, optionally backbone N-methylated macrocycles by the action of a prolyl endopeptidase belonging to the PopB family and N-methyltransferases belonging to the omphalotin methyltransferase family. An exemplary list of prolyl endopeptidases is shown in Table 2. An exemplary list of N-methyltransferases is shown in Table 3.

(99) TABLE-US-00002 TABLE 2 Amino acid sequences of prolyl endopeptidase type cyclizing enzymes *Galerina* SEQ ID

MSSVTWAPGNYPSTRSDHVD TYQSASKGEVPVPDPYQWLEESTDEV DKW *marginata*
NO.: 42 TTAQADLAQSYLDQNADIQKLA EKFRASRNYAKFSAPTLLDDGHWFYFN
CBS 339.88 KDR68475.1

RGLQSQSVLYRSKEPALPDFSKGDDNVGDVFFDPNVLAADGSAGMVLCKF hypothetical
SDGKFFAYAVSHLGGDYSTIYVRSTSSPLSQASVAQGVDGRLSDEVKWF protein
KFSTIIWTKDSKGFLYQRYPARERHEGTRSDRNAMMCYHKVGTTQEEDII
GALMADRAFT_78538

VYQDNEHPEWIYGADTSEDGKYLYLYQFKDTSKKNLLWVAELDEDGVKSG
IHWKVVNEYAADYNIITNHGSLVYIKTNLNAPQYKVITIDLSKDEPEIR
DFIPEEKDAKLAQVNCANEEYFVAIYKRNVKDEIYLYSKAGVQLTRLAPD
FVGAASIANRQKQTHFFLTLSGFNTPGTIARYDFTAPETQRF SILRTTKV
NELDPDDFESTQVWYESKDGTKIPMFIVRHKSTKFDGTAAAIQYGYGGFA
TSADPFFSPIILTFLQTYGAIFAVPSIRGGGEFGEEWHKGRRET KVNTF
DDFIAAAQFLVKNKYAAPGKVAINGASNGGLLVMGSIVRAPEGTFGA AVP
EGGVADLLKFHKFTGGQAWISEYGNPSIPEEFDYIYPLSPVHNVRTDKVM
PATLITVNIGDGRV VPMHSFKFIATLQHNVPQNP HPLL IKIDKSWLG HGM
GKPTDKNVKDAADKWGF IARALGLELKTVE *Amanita* SEQ ID

MPPTPWAPHSYPPTRSDHVDVYQSASRGEVPVPDPYQWLEENSNEVDEW *bisporigera*
NO.: 43 TTAQTAFTQGYLDKNADRQKLEEKFRASKDYVKFSAPTLLDSGHWFYFN
ADN19205.1 SGVQSQAVLYRSKKPVLPDFQRGTRKVG EYVFDPNVLSADGTAIMGTCRF
prolyl SPSGEYFAYAVSHLGV DYT IYVRPTSSSLSQAPEAEGGDGRLSDGVKWC
oligopeptidase KFTTITWTKDSKGFLYQRYPA RESLVAKDRDKDAMVCYHRVGTTQLEDII

VQQDKENPDW TYGT DASEDGKYIYLVVYKDASKQNLLWVAEFDKDG VKPE
IPWRKVINEFGADYHVITNHGSLIYVKTNV NAPQYKVVTIDLSTGEPEIR
DFIPEQKDAKLTQVKCVNKG YFVAIYKRNVKDEIYLYSKAGDQLSRLASD
FIGVASITNREKQPHSFLTFSGFNTPGTISR YDFTAPDTQRLSILRTTKL
NGLNADDFESTQVWYKSKDGTKVPMFIVRHKSTKFDGTAPAIQNGYGGFA
ITADPFFSPIMLT FMQTYGAILAVPNIRGGGEFGGEWHKAGRRET KGNTF
DDFIAAAQFLVKNKYAAPGKVAITGASNGGFLVCGSVVRAPEGTFGA AVS
EGGVADLLKF NKFTGGMAWTSEYGNPFIKEDFDFVQALSPVHNVPKDRVL
PATLLMTNAGDDR VPMHSLKFVANLQYNVPQNP HPLLIRVDKSWLG HGF
GKTTDKHTKDAADKWSFVAQSLGLEWKTVD *Hypsizygus* SEQ ID

MAISPTPWTPNTPYSTRYSHYADPYQWLEENTTEED *marmoreus*]
NO.: 44 KWTTAQEEFTRSYLDKNTDRQRLEDAFRTSTDYAKFSSPTLYEDGRWYWF
KYQ30898.1 YNSGLQPQPLIYRSKGKTLPDFSQDDNVVGEVFFDPNLLSDDGTAALSIY
Prolyl DFSDCGKYFAYGISFSGSDFSTIYVRSTESPLAKKNSGSTDDRLSDEIK
endopeptidase HVKFSAVTWTKDSKGFFYQRYPAHENAKEGIETGGDVDAMIYYHVIGTSQ
SEDILVHSDKSNPEWMWSIDITEDGKYLILYTMKDSSRKNLMWIAELSKN
EIGPNIQWNKIIDVFDAEYHLITNDGPILYVKTNADAPQYKLVMTDISGD
KDISRDLIPEDKNANLVQVDCVNRDTFAVIYKRNVKDEIYLYSKTGIQLS
RLASDFVGAASISSREKQPHFFVTMTGFSTPGTVARYDFGAPEEQRWSIY
RSVKVNGLNPDDFESKQVWYESKDGTKIPMFIVRHKATKFDGTAPAIQYG
YGGFSISINPFFSPTILTFLQTYGAVLAVPNIRGGAEFGEDWHKAGTREK
KGNVFDDFVAATQYLVKNKYAGEGKVAINGGSNGGLLVGACINRAPEGTF
GAAVAEGVMDLLKFSKFTIGKAWTSDYGDPDDPKDFDFICPLSPLHNIP
TDRVLPPTMLLTADHDDRVPVPMHSFKHAATLQYTLPHNPHPLVIRIDKKA
GHGAGKSTEKRIKESADKWGFVAQSLGLVWQEP *Conocybe* SEQ ID
MPPSTPNEYPPTRRSDDVLTYRSEKNGEVVVPDPYQWLEHNTTEETDKWTT *apala* NO.: 45
AQAAFTRAHLDKNPKRNALEEAFTAANDYAKFSAPQLHDDGRWYWYYNTG ACQ65797.1
LQAQTCLWRTRDDTIPDFSKQLDEDVGEIFFDPNALS KDGTAAALSTYRFS prolyl
RDGKYFAYAIAQSGSDFNTIYVRPTDSPLTKRDESGRDP SRLADEVKFVK oligopeptidase
FSGITWAPNSEGFFYQRYPHIDGATLEEGGIATRDLHAMVYYHRVGTPQ
SEDILHRDPANPEWMFGVNVTDNGEYIELYISKDSSRKNMLWVANFAMN
KIGEQQWRKVINDFAAEYDVITNHGPVYYFRTDDGAPKHKILSINIDTN
ERKLLVPESEDAALFSTVCVNKNYMALIYKRNVKDEVHLYTLEGKPVRR
AEDFVGACTISGKEKQPWFFVTMSGFTSPSTVGRYNFQIPEEENRWSIFR
AAIKNLNPNDFEASQVWYKSKDGTNVP MFIVRHKSTQFDGTAPALQYGY
GGFSISIDPFFSASILTFLKVYGAILVVP SIRGGNEFGEEWHRGGMKQNK
VNCFDDFIAATNHLVEHKYAAPGKVAINGGSNGGLLVAACINRAPEGTFG
AAIAEVGVHDMLKFHKFTIGKAWTSDYGNPDDPHDFDYIYPISPVHNVP
DKILPPTLLLTADHDDRVPVPMHTFKLAATLQHTLPHNPHPLLLRVDKKAG
HGAGKPLQLKIREQADKWGFVAQSFQLVWRDGV *Amanita* SEQ ID
MPPTPWAPHSYPPTRRS DHVDVYQSASRGEVPVPDPYQWLEENSNEVDEW *bisporigera*
NO.: 46 TTAQTAFTQGYLDKNADRQKLEEKFRASKDYVKFSAPTLLDSGHWYWFYN
GenBank SGVQSQAVLYRSKKPVLPDFQRGTRKVGVEYFDPNVLSADGTAIMGTCRF
HQ225841.1 SPSGEYFAYAVSHLGVDYFTIYVRPTSSSLSPAPEAEGGDGRLSDGVKWC
POPB KFTTITWTKDSKGFLYQRYPARESLVAKDRDKDAMVCYHRVGTTQLEDII
VQQDKENPDWTYGTDASEDGKYIYLVVYKDASKQNLLWVAEFDKDGVKPE
IPWRKVINEFGADYHVITNHGSLIYVKTNVNAPQYKVVTIDLSTGEPEIR
DFIPEQKDAKLTQVKCVNKG YFVAIYKRNVKDEIYLYSKAGDQLSRLASD
FIGVASITNREKQPHSFLTFSGFNTPGTISR YDFTAPDTQRLSILRTTKL
NGLNADDFESTQVWYKSKDGTKVPMFIVRHKSTKFDGTAPAIQNGYGGFA
ITADPFFSPIMLTFMQTYGAILAVPNIRGGGEFGGEWHKAGRRET KGNTF
DDFIAAAQFLVKNKYAAPGKVAITGASNGGFLVCGSVVRAPEGTFGA AVS
EGGVADLLKFNKFTGGMAWTSEYGNPFIKEDFDFVQALSPVHNVPKDRV
PATLLMTNAGDDRVPVPMHSLKFVANLQYNVPQNP HPLLIRVDKSWLG HGF
GKTTDKHTKDAADKWSFVAQSLGLEWKTVD *Lentinula* SEQ ID
MFSATQESPTMSVPQWDPYPPVSRDETSAITYQSKLCGSVTVRDPYSALE *edodes* NO.: 47
VPFDDSEETKAFVHAQRKFARTYLDEIPDRETWLQTLKESWNYRRFTV PK GenBank
RESDG YTYFEYNDGLQSQMSLRVKVSEEDILTESGPGGELFFDPNLLS GAW09065.1
LDGNAALTGSMMSPCGKYWAYGVSEHGSDWMTTYVRKTSSPHMPSQEKGK The DOE
DPGRMDDVIRYSRFFIVYWSSDSKGFFYSRYPPEDDEGKGNTPAQNCMVY Joint

YHRLGEKQELQEDPEHPFWLWALQLSPSGRYALLTASRDASHTQLA Genome
KIADIGTSDIQNGIQWLTIHDQWQARFVIIGDDDDSTIYFMTNLEAKNYLV Institute
ATLDIRHSEAGVKTLVAENPDALLISASILSTDKLVLYLHNARHEIHVH (JGI)
DLNTGKPIRQIFDNLIGQFSLSGRRDDNDMFVFHSGFTSPGTIYRRLNE 011197;
DSNKGTLFRAVQVPGLNLSDFTTESVFYPSKDGTPIHMFITRLKDTVPVDG LENE_011197)
TAPVYIYGYYGGFALAMLPTFSVSTLLFCKIYRAMYVVPNIRGGSEFGESW
HREGMLDKKQNVFDDFNAATKWLVAANKYANKYNVAIRGGSNGGVLTTACA
NQAPELYRCVITIGGIIDMLRFPKFTFGALWRSEYGDPEDEDFDFIYKY
SPYHNIPSGDVVLPAMLFFTAAYDDRVSPLHSFKHVAALQYNFPNGPNPV
LMRIDLNTGHFAGKSTQKMLEETADEYRCDLLCCNLQL *Omphalotacae* SEQ ID
MSFPGWGPYPPVERDETSITYSSKLHGSVTVRDPYSQLEVPFEDSEETK *olearis* NO.: 48
AFVHSQRKFARTYLDENPDREAWLETLLKSWNYRRFSALKPESDGHYYFE The DOE
YNDGLQSQLSLYRVRMGEEDTVLTESGPGGELFFNPNNLLSLDGNAALTGF Joint
VMSPCGNYWAYGVSEHGSDWMSIYVRKTSSPHLPSQERGKDPGRMNDKIR Genome
HVRFFIVSWTSDSKGFFYSRYPPEDDEGKGNAPAMNCMVYYHRIGEDQES Institute
DVLVHEDPEHPFWISSVQLTPSGRYILFAASRDASHTQLVKIADLHENDI (JGI) 2090;
GTNMKWKNLHDPWEARFTIVGDEGSKIYFMTNLKAKNYKVATFDANHPDE
OMPOL1_2090 GLTTLIAEDPNAFLVSASIIHAQDKLLLVYLRNASHEIHIRDLTTGKPLGR
IFEDLLGQFMVSGRRQDNDIFVLFSFLSPGTVYRYTFGEEKGHSSLFRA
ISIPGLNLDDFMTESVFYPSKDGTSMHMFITRPKDVLLDGTSPVLQYGYG
GFSLAMLPTFSLSTLLFCKIYRAIYAIPNIRGGSEYGESWHREGMLDKKQ
NVFDDFNAATEWLIANKYASKDRIAIRGGSNGGVLTTACANQAPGLYRCV
ITIEGIIDMLRFPKFTFGASWRSEYGDPEDEDFDFIFKYSPYHNIPPPG
DTIMPAMLFFTAAYDDRVSPLHTFKHVAALQHNFPKGPNPCLMRIDLNSG
HFAGKSTQEMLEETADEYRLKVQ

(100) TABLE-US-00003 TABLE 3 Amino acid sequences of N-methyltransferases
Lentinula SEQ ID

METPTLNKSGSLTIVGTGIESIGQMTLQTL SYIEAADKV FYCVIDPATEAF *edodes* NO.: 49
ILTKNKDCVDLYQYYDNGKSRMDTYTQMSEV MLREV RKGLDVVG VFYGH PG GenBank
V FVNPSLRALAI AKSEGFKARMLPGVSAEDCLYADLCIDPSNPGCLTYEAS GAW09067.1
DFLIRERPTNIYSHFILFQVGC VGIADFNFTGFENSKFGILVDRLEKEYGA The DOE
EHPVVHYIAAMLPHEDPVT DQWTIGQLREPEFYKRVGGVSTFYIPPKERKE Joint
INVDIIRELKF LPEGKVPDTRTQIYPPNQWEPEVPTVPAYGSNEHAAIAQL Genome
DTHTPPEQYQPLATSKAMTDVMTKLALDPKALAEYKADHRAFAQSVPDLTA Institute
NERTALEIGDSWAFRCAMKEMPISLLDNAKQSMEEASEQGFPWIIVVG VVG (JGI)
VVGSVVSSA 011194; LENE_011194) *Omphalotacae* SEQ ID
METSTQTKAGSLTIVGTGIESIGQMTLQALSYIEAAAKV FYCVIDPATEAF *olearis* NO.: 50
ILTKNKNCVDLYQYYDNGKSRMLNTYTQMSELMVREV RKGLDVVG VFYGH PG The DOE
V FVNPSHRALAI AKSEGYRARM LPGVSAEDCLFADLCIDPSNPGCLTYEAS Joint
DFLIRDRPVSIHSHLVLFQVGC VGIADFNFTGFDNNKFGVLVDRLEQEYGA Genome
EHPVVHYIAAMMPHQDPVTDKYTVAQLREPEIAKRVGGVSTFYIPPKARKA Institute
SNLDIIRRELLPAGQVPDKKARIYPANQWEPDVPEVEPYRPSDQAAIAQL (JGI) 2087;
ADHAPPEQYQPLATSKAMSDVMTKLALDPKALADYKADHRAFAQSVPDLTP
OMPOL1_2087

QERAALELGDSWAIRCAMKNMPSSLLDAARESGEEASQNGFPWVIVVG VIG VIGSVMSTE
Dendrothele SEQ ID

MESSTQTKPGSLIVGTGIESIGQMTLQALSYIEAASKV FYCVIDPATEAF *bispora* NO.: 51
ILTKNKNCVDLYQYYDNGKSRMDTYTQMAELMLKEVRNGLDVVG VFYGH PG The DOE
V FVNPSHRALAI ARSEGYQARMLPGVSAEDCLFADLCIDPSNPGCLTYEAS Joint
DELIRERPVNVHSHLILFQVGC VGIADFNFSGFDNSKFTILVDRLEQEYGP Genome

DHTVHYIAAPMHPQDPTMDKFTIGQLREPEIAKRVGGVSTFYIPPKARKD Institute
 INTDIIRLLEFLPAGKVPDKHTQIYPPNQWEPDVPTLPPYGQNEQAATRL (JGI)
 EAHAPPEEYQPLATSKAMTDMTKLALDPKALAEYKADHRAFAQSVPDLTP 765759
 QERAALELGDSWAIRCAMKNMPSSLLEAASQSVEEASMNGFPWVIVTGIVG VIGSVVSSA
Rhizopogon SEQ ID
 MTTDTKRGTLTIAAGSGIASIAHITLETLSYIKESDKLFYLVCDPVTEAFIQ *vinicolor* NO.: 52
 DNATGDFFDLSVFYDKNKSRYDSYIQMCEIMLRVRAGHSVLGIFYGHPGV GenBank
 FVSPSHRAIAVAREEGYKARMLPGVSAEDYMFADLEFDPSQSTCNTYEATE AM-OR11-
 LLLRDRPLDPAIQNIIWQVGSVGVVDMEFEKSKFHLLVDRLEQDFGPDHKV 026;
 VHYIGAVLPQSTTTMDIFTISDLRKENVAKQFGTISTLYIPPRDEGPVSSS OAX31299.1
 MTQAFDFKAGAMVYSPVKWAGPKLNIVSALSPYERDVISQIDTHVAPEGYK
 ILHTSAAMNKFMTDLSLKPKFLEEYKLYPEAVVESAEGLSNLEKFGKFGS
 DGAVYILMKATESDIASGRQLTEDEIAKAHKSVMGFPTVLVILPTVIVVLIG RE *Rhizopogon*
 SEQ ID MSTKRGTLTIAAGSGIASVGHITLGTLSYIKESDKIFYLVCDPVTEAFIYDN *vinicolor*
 NO.: 53 STADCFDLSVFYDKTKGRYDSYIQMCEVMLKAVRAGHDVLGVFYGHPGVFV
 GenBank SPSHRAIAVARQEGYKAKMLPGISAEDYMFADLEFDPSVSGCKTCEATEIL AM-
 OR11- LRDKPLDPTIQNIIWQVGSVGVVDMFEFSKSKFQLLVDRLEKDFGPDHKVVH 026;
 YIGAVLPQSTTTMDTFTIADLRKEDVAKQFGTISTLYIPPRDEGHVNLSMA OAX32862.1
 KVFGGPGASVKLNSIKWAGPKLNIVSANDPHERDVIAQVDTHVAPEGHKK
 LRVSAAMKKFMTDLALKPKFLEEYKLPVAVVESAEGLSNLERFGLKFARS
 GPADALMKATESDIASGRQLTEEEIAQGTGPVGLQTALALLVLLGLGVAIV TRPDD
 (101) In other embodiments, the cyclization comprises reacting with beta-lactamase (FIG. 12). As
 shown in FIG. 12, a variable region is excised and end-to-end cyclized by the actions of an N-
 methyltransferase and a beta-lactamase family member. Table 4 shows an exemplary list of
 lactamase and amino acid sequences of the processed cyclic peptides. In some embodiments, some
 of the sidechains of the randomized residues are subsequently isomerized from the L- to D-
 configuration or decorated with additional modifications like hydroxylation, halogenation,
 glycosylation, acylation, phosphorylation, methylation, and acetylation.
 (102) TABLE-US-00004 TABLE 4 Amino acid sequences of the N-methyltransferase
 and beta-lactamase processed cyclic peptides *Rhizophogun* SEQ ID
 MAKVFGLVLGFLSQTFTYPSQVWFSPVGANNGQVITPELSNSIQETLDVWN *vinicolor*
 NO.: 54 ITGLSVAIIPKSGEPEYHSWGDRTEDGESVTQDTL FHMASVSKAFCVSA LG
 GenBank ILMDDFEHGRNVTPLPPALTEFNWHTSIQDLLPGEWQLMDEWASRKANMKD
 OAX32863.1 ILSHVSGLP RHDFAFGPYESPKEAVSRLRYLRPAFELREQWSYNNQMFMVA
 hypothetical GHIVETYS GKTYTSFVEDRIFTPLGMSSSTESPAKAAKTGKFTQGWTSSGR
 protein beta-
 LLPELFPEDMVMLMAGAGGVISSAVDMSKWVALWLNKGVYDNTVTVIPSSVY lactamase
 GNASQSYAVSISTPVDSEHSIQGYGLGW FQNSYLGHNVVYHSGSIPGLSML (transpeptidase)
 VSFLPDDDDVG FVVFANGGDKAAPVMNISNSIIDAALHLRSGPAPPIMPEKK
 AVTSPSEDIVNLELPLEEFSGT YTDPGYGTFTFCSPSSSSSYCQQVMTDFT
 AVDSVHPSAPSPLQLLA AWP RMGSSHRAVHQSGNKFLLLCTALFPEGYGR
 DSTPFETA EIGTPGATAEFVVEDGKVVGEGLEGLVDQVTERERTQTTVKDR AEVWFDKV
Rhizophogun SEQ ID
 MIMAKVFGLVLGFLSQTFTYPSQIRLSPVGVNNGQVITPELSNSIQETLDV *vinicolor* NO.:
 55 WNITGLSVAIIPKSGEPEYHSWGDRTEDGESVTQDTL FHMASVSKAFCVSA GenBank
 LGILMDDFEHGRNVTPLPPALTEFNWHTSIQDLLPGEWQLMDEWASRKANV OAX34183.1
 KDILSHVSGLP SHHFAFGPYESPKEVVSRLRYLRPAFELREQWSYNNQMFT hypothetical
 VAGHIVETYS GKTYTSFVEDRIFTPLGMFSSTFSPAKAVKTGKFTQGWTSS protein beta-
 GRLLPEFFQEDMIMPMAGPGGVISSAVDMSKWVALWLNKGVHDNTIIPSS lactamase
 VYGNASQSYAVSISTPVDSEHSILGYGLGWFRNSYLGHDVVYHSGSIPGLS (transpeptidase)

TLVFLDDVDFVFANGDNKAAPVMNISNRIIDAALHLRSGPAPPIMPE
 KKA VTSPSE DIVNLELPLEEFSGTYTDPGYGTFTFCSPSSSSPYCQQVIAN
 FTTVDSVRPSAPSSLQLLAAWPRVGSSHIRTVHQSGNKFMLLPTALFPEGY
 GRDSTPFETA EIGTRGAPVEFVVEDGRVVGFGLFGLVGQVTERERTQTTVK
 DRAGVWF DKV *Rhizophogun* SEQ ID
 MSTKRGT LT IAGSGIASVGHITLGTLSYIKESDKIFYLVCDPVTEAFIYDN *vinicolor* NO.: 56
 STADCFDL SVFYDKTKGRYDSYIQMCEVMLKAVRAGHDVLGVFYGHPGVFV GenBank
 SPSHRAIAVARQEGYKAKMLPGISAEDYMFADLEFDPSVSGCKTCEATEIL OAX32862.1
 LRDKPLDPTIQNIWQVGSVGVVDMEFSKSKFQLLVDRLEKDFGPDHKVVH hypothetical
 YIGAVLPQSTTTMDTFTIADLRKEDVAKQFGTISTLYIPPRDEGHVNL SMA N-
 KVFGGPGASVKLND SIKWAGPKLNIVSANDPHERDVIAQVDTHVAPEGHKK
 methyltransferase
 LRVSAAMKKFMTDLALPKPFLEEYKLPVAVVESAEGLSNLERFGLKFARS
 GPADALMKATESDIASGRQLTEEEIAQGTGPVGLQTALALLVLLGLGVAIV TRPDD
Rhizophogun SEQ ID
 MTSDNLQPEVISANWLKSLEAASSTGDTASFVSHFLPDGWFRDMLCFTWNF *vinicolor*
 NO.: 57 RTL SGQEKIHGFISEVVDGQSRLSYSHLHDFKLDDHSVNAPSPFKLPGPPD
 GenBank IEGVQGAFTFSITKPAAYGRGFFRLTQDVHGNWKALTFTNMQDLVGHEES
 OAX34185.1 SADEYDPHEKANPTVVIVIKVGGGQSGLICAARLGKLGIRALVIDKNARVG
 hypothetical DIWRQRYAEALPSFAVLSRQETQVPEPYAAYSQISKLLPYPSNFPKYLPKG
 protein KLANFLESYAINQELCIWLSSTVSPSPVYDSFSARWTVEVEHENRKVILHP
 FAD/NAD(P)-
 KHLVLATGHGRPRIPTWNGMDDFQGTLYHSDFHRDAEKFRGKCVVVIGAGN dependent
 ASGDICEDFVAQGAAEVTIVQRSATCVVSSATADAFVFKLPFSDKTPIEEL oxidoreductase,
 DFRHNSMPLAFVLQLMKSGGTQHMKAHDKHEHGLRKAGFNLTWEPSPGSG D-amino
 acid EVGLLGFVFERAGSGTMDTGFGKLIVEGTVKVKQGQNISHFDKEGITFKD
 dehydrogenase GSKLPADVIVAATGNELTMDAIRAVLGDTIAEQLPPK VWGLDAEGELNQMY
 RPSGHPGLWFAVGSLGMTRFCSKHLGLQILAQEVGIA
 (103) In some embodiments, the cyclization comprises reacting with a prolyl endopeptidase, an N-
 methyltransferase, and a hydroxylase (FIG. 15). In some embodiments, the bicyclization comprises
 further modification of the indicated anchored residues on the cyclized peptide, forming an internal
 tryptathionine bridge (FIG. 16). FIG. 17 illustrates a biochemical steps to create the tryptathionine
 bridge with hydroxylase and dehydratase. Step 1 involves hydroxylation of the 2-position of the
 indole ring of the tryptophan residue by a hydroxylase belonging to the cytochrome P450 family of
 oxygenases (FIG. 17). An example of such hydroxylase is shown in TABLE 5.
 (104) TABLE-US-00005 TABLE 5 Amino acid sequences of a hydroxylase *Galerina*
 SEQ ID MGKMAYHTVLDDIALYLLGSAALVIFYRSFFYPYFLSGRRLAPGPTKGELS
marginata NO.: 58
 KELKQFNNEINVHFLRHMVKEYGPIFRLVGAPMIPGPGLVVCTPTAQQRIL CBS 339.88
 KDR84981.1 VSNSINYGQPRLAFFRWVTGGLFTLPEREHRGMRKILDPVFSFRNLSTTG
 hypothetical VYYNTVQSLITIFRSKIDGENGAKDGDVILVYEWLARLAIDNVSEAILGFK
 protein LDTLHDPNNELITTLDEL SRIPTAAFELLVRVP GFRLRLVTFDSVRHSTLWQ
 GALMADRAFT_260690
 RRVPGRLGVFFTFMRCLSTIRKNALAIKATILQEDSANRDLNVISVLQHMQ
 SSDETANADIAGNIIMLWMSGRATIATRISWLLWLLAKDQQCQQQLRDEIA
 PLFSRDP RP DY RSLDKLQWLDSVIMESIRLFLFGPNIRVALNDDYIDGVFV
 PKGTVVVIPLDLFTRGDIWGEDPDQFKPARWLDSTKRYKISPPFLSFLTGP
 HRCIAKGMAIMQTKIVIASLIANFEFKPAYEGQHVEGNPSIIGHGMPLHVK PIRPS

(105) Step 2 involves the formation of a tryptathionine bridge between the 2'-hydroxyl position on
 tryptophan and the thiol group from the cysteine residue. This condensation reaction is catalyzed

by a novel family of dehydratases. Examples of the dehydratases are shown in TABLE 6.
(106) TABLE-US-00006 TABLE 6 Amino acid sequences of dehydratases *Galerina*
SEQ ID MPYVPDPKYFEHREQSSGATLYYCLVCRDGRERQPHHIKTHEASQAHRTAL
marginata NO.: 59

SVFDSQAESSSQTHGNPTQPGYFDPVIDDAVRALLVSGSGDPHQPLYPAG CBS 339.88
KDR80488.1 HPNVYGEPNFTDSRRRTSPVTGIDWDQFEAQEDTHAVPSAQDQLRADICQA
hypothetical TLDWLNDDISDDDEREPSEVDSVDSDAESDREPIPDDQPRKRARTNRDNPI
protein SEDWYPWQDKITCTLDILMHLPRSVFSRKQLDLFLWLLRVNNVDDVPTGKS
GALMADRAFT_136963

MKMLNKILQGMCGIETIAYEGKLGHNHYHVNNIAQILAQELCNPKVGPHIYF
YPEDSGDNLAEARQAARWLHEL RPEETTPMIHLPSGDYYIYEPAMLSNRSF
CIPFRWFTRNGKFHARAWSLETGVVDNTLGWIVHKENEVEISEDDLLKDFT
RFSSDCEAYNVPHPSRILGVSCADSGNLLPWNHTNPVLGNRWRQLAKGHRT
LCLPLWMYCDDTSGNTSKKWNEHNSFLFTLAGLPREHTAKEYNIHFLCTSN
LAPPLEMMDGVVVSQIEAAQQNGIWAWDCVRKEPVLIFPTILALLGDNPMHS
EFACHIGLRGKFFCRTCWVKGSDAQDDANIVTPGLHETPENSAPSPAPSP
APSPAPSPAPSPALSMAPQSQPPTPSEPSMQVPAPPSTAAPT KARGKKKET
MSAMLNRITAFIKPGRLRNKSETQKTLQNFKEQAQTIGAKTKLKTARTETG
IKDTVQEFFFEKLFSSYKNKRGPQAKQEALDQAVNQLPSDITSPVWRLKGL
DPHQDTPVEILHVVL LGFIKYFWRDLVQNQINDDQKQTLIQRLNSFDVTGL
GITQLGGETLVNYAGSLTGRDFRAVAQVAPFVIYDMVPADVFDWLALSKL
VPLVWQPYIENVAQYLTTLEHEIHVFLRTARWTTGWFNKSKFHIILHLPS
HIRRFGPAILFATEAFESFNAVIRAKSVHSNRQAPSRDIALAFAQGNRIRH
LLSGGHFLSADTHMVVDPDQPQLGQYERLARGRWRSVGP GPGHLVSAEPIL
PSYLGIPPQSTTSSAGLCKRTKTPPQTFLQTLTGLKLPNVSRPGARELWQT
CSEVYLLNDDKCLIGHHVIVQRQSEQASFVSPFFIARIGEILQKVG SANHA
HDKPDGILVQTLKSSEVADKFQMPRLVPQNEWSFVPLADILCTVNAQHDCD
RNGCTASGFYVYQERIQTNDQRPVVEHV NQPEDFILNTAQMRDALHLQKF
RIRSRLDEQTIHESVARTINQRKAQDNSSSGTGGAGVSGRGRGRGRGRG
GGVEGPSTSRGRGGGIEGRGASSSSGNRGRGRGARSAQSVPF

Galerina SEQ ID
MPRKKPAPECFETDEASKMIRCLICKENDTVQQGTWIKHGSASQHIETNAH *marginata*
NO.: 60 KLA VARREQLLQVQQEEERRLQEIYGGNTIPLSGNAQLYPTYPRANMYGNQ
CBS 339.88 KDR74877.1

DAVDTDMDNQNSPPQAYMLCDADIPDLGIKPIERPDP SQRERLRQQVEQL hypothetical
LLQAEHEDEFGSPDDPDDLSTNIAQAFADLDLEEMLDEEEVEDYFNQVSP protein
EHDYYPYPNKTTMLLDILDNL PRLRMSSNQLRLILWLLKQTGVS NVPSFSG
GALMADRAFT_99137

FRNMQTHLRNMCGTTPKQHVSSLGNIFYSN NIGESVMRDFANPEVAKHLHL
YPEETEGPISEVWQAERWKEFAPSELTPMFSQGHRQFFIDEVAQLQDGQYV
IPRNWVMRK GKLTSDCHIVTVNPVRFSKLHGSLVLVLKQCFQSGWTLLSET
QIFHADDFQFNYFDVVSRI RGPISWSEGTEVPAMPNNLREL AGDDDLVVIM
VPLWCDDVSGNKS KQYNKHINVYMAN SNIPGRLLQQEYFVRFVSTSPNATS
PEQFSALKDQINETQKKPIQCYN AHTNKKTRAILRVPGLPADNPQQSEESC
HMGGNANCKCRKCHVGGPHEKKESNEGYHEHYLTGIKRS AEETRLELEKQI
KLAMYGVEKPINETQTN TGTDKVAQHWIDILLAKSRELKSANPSRSVEEI
AQELQTW FDEQPGDKINPLLSIAGLDPTQDTPVEILHTILLGIVKYAWHHL
HSNWTEAEQNLFTVRLQSTDIDGLSV PPIRVAYMMQYRNGLIGKHFKTLMQ
TLPFHVHGTVSDAQFKLVKAIGELGSVLWVHEIGDMEKYLS DLEILIGNVL
DAFAEIDPSTAMYARFIYEPMPVPSKIIVKLKLHMLPHLIEDIKRFGPAIR
NSTEVFECFNAIFRLCSILSNHQAASRDIALKFASMDRLKHMLSGGYWLSE

VEEGKFEWIRAGENVIRNQSEPTIQRHLGWAPSAKFQSGRKRTPTPTSWEN
TKASQFMDSEETAAGFPNPRLLSWRKGVTTTAQSGDRCSTGSWVVARNHK
VCYILASHYCSIAKNDQGESCIGRIHEIIGPDEKSASSTGIITLECFQLGK
EHHPDFGLPTLQRPQADLPKYILKAWQDPLFIFSAHHDCHTASCQATALQP
QLQERQLTSRMNKLIAHNDSDFIINLYGLHNAILLREFLPRELTAPQPLH
QDRKAFHYEVA AKLRVQQA EKRAKTNARRKATRAANKAKQVERQKQNP DHE
QESEQEMDERPNSENGSDIELGGDDIEVETRRKRRRN *Hypsizygus* SEQ ID
MGRRAEELPAYVELSEDGTLVRCNLCLMHNRLDYSKEWIRQKGWRSHKGS *G marmoreus*
NO.: 61 IHDRSEAKQRVLDDAAMD LQEPASAEVEVVT FNDILIINAPKTPTGNMQSE
KYQ37095.1

EQAMWDHFDAGSFTLEAGEDPNHSSQRLYQDLARKADAYGAWDGTEALPEY hypothetical
RDLDDVSQFLDEDEEEDLLSEILRGLGLEEEHEDSSDRNP AEELNSPWYPY protein
GSKLMFLLDTIDNLPRLRISGAMMRVFLWLLREVGVRQVPSFDKLRKIQRK Hypma_08924
LREGSGVPTVHWMSPKGNAYSFN DPAVIVANDWAS PITRPHLRYPVIPKD
GVITEVYHAEKWHREINRHFLT PMYDDGFRHYFIDELAQLKDGRYAVPVRW
LEDVDGRIVADAWRVELEDDNRATIIDTATVRIHSQELALNFEEIIESNLM
PEWSDTTTEAGHPSRMPNPDRALAE GDP IYTSFIDIFGDDVSGNRSKSWNK
HWNMYISHRNLP RKLLHQQYH THFVSTSTFASIPEQFVG VKEA IESTH SKP
VKVRDADTGKQIRLKIYCNC GPGDNPSQSETSGHIGGNGNYP CRKCHTGGT
QKSKETDEGFYKMFTAGEARSSKETLAEVKSQVEAACTGVAKTVADAQSDT
GVKDAYTQYWIDAIEKARAMQKENPGMPTTTIQATLIKWVYDHEEAIYNS
FLTLDGFDASRDTPVEILHTILLGIVKYLWHSHTSWNAAQKKIYSTR LQG
TNTQGLSIHHIRANYIMQYANSLIGRQLKT LAQVNVFHVYDLVDPLRFLFT
KATGELCALLWFTEIRDLEEYLSDVDIAAANVLDIAAVIDPSKIVSKIYH
LLSHLREDIIRFGPLVGVATEVFECFNAVERYCSILSNHLAPSRDIAYKLA
AQETMKHFLSGGWWHV KDSVDLQGNPKWVQPGPSVRTFMASNPVLHTLCGW
TRNNDSTPGTVKSEPRKRGPD KQTLPLVRLAWLETQGSRALNNTSPNNET
QWQRCKYVIAETQDQC NVGSWVFARSPLLENIPGRIVEILQDTSASPSA
FVVIDVFQVSATRDEVEGMPVLLRRFNECCLHVIPASSVIFDFNAQHDCRY
AKCEATGEQPLIQERVPSGV TENFVVHKAIDRYLINIHALHNAHLIRATLP
RDLTAPIPYAPNREAHHSAIAAELRSAQDTKRAKTA AKTAANAAAKKAEAA
LKDTTSGPA AKRRRV DDEGS GEEDNRD VDMVSV *Galerina* SEQ ID
MAKGRKLNNPLPDFIEISNDGLQVRCTLCLAA RQHNGSGWIKRGSVSNHLK *marginata*
NO.: 62 SDNHTNSLEAH EMKKSAEKA EGRSVQEEIAMEEGMDFVILSSKI QPEITAP
CBS 339.88 KDR73903.1

ARAPRRSNEEQEMWD RYTLGGEVFDAGVDHTLVEAEERKRLEREATDFDLW hypothetical
HGADFLPEEDPNDGELLLDELEQDDILSELLRNAHLNAPDAADVLT EEPRA protein
AADPRICDAWSPYESKMMFLLDTLDNLPRLRISNSLMNVFLWILREGGARD
GALMADRAFT_141673
VPSLYHLRQVQTTLRKSTGVPTTQHKSPKGNVYSMNDPRTL VAMDWANPVI
CDHIRRYPVIPRNGVISEVYHAQKWRKD VDPHTLSPMYDAGNCHYYIDEVA
RLKNGTFIIPVRWLEDED RNV CADAYVVQFDDQFIASVVDGETIIVQASDL
QNNFLDLKDMGLLPTWGNQTIESGHPARMPNPDRALAE GDP LYTSWIDVFG
DDVSGNRSKNWNKHWN IYISHRNLP RKLLQQEFH THFVSTSPVASVTEQFH
GIKQVIELTHKSPVKVRHGTSGAQIRFKINVNCGPGDNPAQSEVCGHIGVN
GNKLCRKCHTGGTHEVKESDEGFNSLFEPGDARSAQEIVADVESQVQLACL
GIAQHVNQQTKNGIKDAYTQYWIDY LINRARTLRKEQPRRTTADIQSELL
VWVQEHKDEIYNPFLKLDGFDAAVDTPVEILHTILLGIVKYLWHGSHTSWT
AIQKQTYSVRLQSTDTSGLSIHAIRANYIMQYANSLIGRQFKTIAQVNVFH
VYDLVDTTQFLLTKAVGELTALLWIPEIANMEEYLLDVEAAAANVLDL FAL

IDPSKMTKLKLHLVHLKADILRFGLVGVATETFECFNAIFRFSIYSN
HLAPSRDIAFQLASQEVLYKYRLTGGWWPASDGEWKRPGPSVRNFIHDHPTL
QALLGWTKEEKLNGSFRLEPLKRDASQKIESRKHLPWLQTQGAKAVNSSE
DNDSKWTACRFVANSGBKCSVGSWVFATSPFNSNQSVTGRIVEVLAESG
KRAVVVLDIFEVCSTRHKIFGMPMLARRHEEPVYAVIASTNIEFLYNVQHD
CPLAKCTASGKQPLIQERVESGLFKTYIEHKPIEREVINTHAFHNAHRLRA
VLQRSVLVPIPLYPPEIRKTKHAEFAHNLQATQKVKLEARAAQKAKEIITP
ADKTDSTIPKKRTRSEMETETDDTAIATQADVFFNAQGCP

(107) Step 3 describes S-oxygenation of the tryptathionine thiol by a flavin-monooxygenase enzyme that converts it to a sulfinyl form. Examples of such monooxygenase are shown in TABLE 7. Step 4 describes potential future modification steps such as hydroxylation of side chains on the peptide such as the hydroxylation of position 6 on the indole ring of the tryptathionine-forming tryptophan residue by a P450 family monooxygenase.

(108) TABLE-US-00007 TABLE 7 Amino acid sequences of monooxygenases *Galerina*
SEQ ID MVQIKRLLLGFLSSPSQTPLESNHGPVPSKSIYVVGAGSAGLAMRLTLVEL
marginata NO.: 63

EAFSRNNWEVVLYEERESVGGIWLPDNDVFPPEIPKTPLYPLLRTNTPVP CBS 339.88
KDR68385.1 SMTYPGFPPFPSTPLYPRHDHVEAYHLRYARRHNLLDFIKFDTMVEKAFWN
hypothetical GTPEEGYWNLTLSKEGRMRYKTEDHLVVATGNNHIPHIPVWKGQEDWLAS
protein PANHSRKIIHSVYYRGPEAFSNQTVLIVGNGGSGRDAATQILGYASQTFMS
GALMADRAFT_104945
IRRSYGPVDDGVIVKPDISHFTEAGVVEVDGTILDPDVILLGTGYEMQKPL
LSEGGEISFDPTAKDNSSVRGILVTNGHYIFPLHRHIFSLSPRYPPNALAF
IGLLSFIASCPSDIAQSLFAAHAILDPSILPPRHLLLEELASYEDKARRQG
LDPYLKGPIMLNNTSNDYQDELVEYLKQKNAIPDDGKKFVEEWRREILAYH
YLQRGWSRIEKLGMGPAWTEGVKTEAQWFDLMTRVNEWQKNWETENGIAFR VDLDLTG

(109) A gene organization within two exemplary loci in *Rhizopogon vinicolor* that encode for methyltransferase, beta-lactamase, hopene cyclase, beta-lactamase 2, dehydrogenase, and glycine oxidase is shown in FIG. 13.

(110) The sequence which flanks the encoded random peptide library can be, for example, as shown in FIG. 14, by using N-term and C-term flanks from the MSDIN family genes (toxin preproprotein sequences) identified in the genomes of *Amanita bisporigera* and *Amanita phalloides*. The low consensus central regions indicate areas where a random peptide library could be inserted to facilitate post-translational processing into a cyclic peptide. See Pulman, Jane A., et al. "Expansion and Diversification of the MSDIN Family of Cyclic Peptide Genes in the Poisonous Agarics *Amanita Phalloides* and *A. Bisporigera*." *BMC Genomics*, vol. 17, no. 1, 15 Dec. 2016, p. 1038., doi: 10.1186/s12864-016-3378-7.

(111) The enzymes can additionally be targeted to a specific cellular compartment to increase peptide synthesis efficiency and increase yield for peptide production purposes.

(112) Disclosed herein, in certain embodiments, is a method of detecting interaction between a first test protein and a second test protein in a host cell, comprising: expressing in the host cell a first fusion protein comprising the first test protein and a second fusion protein comprising the second test protein; delivering a first molecule to the host cell; modifying the first molecule while in the host cell via a modifying enzyme; and allowing the first molecule to modulate the interaction between the first test protein and the second test protein, wherein the first molecule is a product of an encoded DNA sequence, wherein the first molecule comprises a randomized polypeptide library and one or more modifying enzymes, wherein the one or more modifying enzymes modify the randomized polypeptide library.

(113) Host Cells

(114) In some embodiments, the host cell is a eukaryote or a prokaryote. In some embodiments, the

host cell is from animal, plant, a fungus, or bacteria. In some embodiments, the fungus is *Aspergillus*, *S. cerevisiae*, or *Pichia pastoris*. In some embodiments, the host cell is a haploid yeast cell. In other embodiments, the host cell is a diploid yeast cell. In some embodiments, the diploid yeast cell is produced by mating a first host cell comprising DNA sequences encoding the first chimeric gene, the second chimeric gene, and the third chimeric gene, to a second host cell comprising DNA sequences encoding the death agent, positive selection reporter, and the mRNA comprising a nucleotide sequence encoding a polypeptide. In some embodiments, the plant is *Nicotiana tabacum* or *Physcomitrella patens*. In some embodiments, the host cell is a sf9 (*Spodoptera frugiperda*) insect cell.

(115) Disclosed herein, in certain embodiments, is a host cell configured to express a first fusion protein comprising a DNA-binding moiety; a second fusion protein comprising a gene activating moiety; a third fusion protein comprising a different DNA-binding moiety; a death agent, wherein the expression of the death agent is under control of a promoter DNA sequence specific for one of the DNA-binding moiety; a positive selection reporter, wherein the expression of the positive reporter is under control of a promoter DNA sequence specific for the other DNA-binding moiety; and a polypeptide of 60 or fewer amino acids, wherein the polypeptide modulates an interaction between the first test protein and the second test protein; wherein the host cell optionally has a mutant background enabling uptake of small molecules; and wherein the host cell optionally has a mutant background enabling increased transformation efficiency.

(116) Disclosed herein, in certain embodiments, is a host cell comprising a plasmid vector which comprises the components of PLASMID 1, or any combination of the components of PLASMID 1; or the plasmid vector, wherein a DNA sequence encoding a first polypeptide is inserted in frame with Gal4-DBD, a second polypeptide is inserted in frame with LexA-DBD, and wherein a DNA sequence encoding a third polypeptide is inserted in frame with VP64-AD.

(117) Disclosed herein, in certain embodiments, is a kit comprising PLASMID 1, PLASMID 2, and PLASMID 3; and transfectable host cells compatible with PLASMIDS 1-3, or any combination thereof. In some embodiments, the provided host cells are already transfected with PLASMID 1 or 2. In some embodiments, the kit includes selectable agents for use with host cells transfected with PLASMIDS 1-3. In some embodiments a library of variants of PLASMID 1 are provided, wherein more than a single pair of Y2H interactors are represented. Such a library can be used to, for example, screen for protein-protein interactions that are inhibited by a defined agent. In some embodiments a library of variants of PLASMID 2 are provided, wherein a plurality of different short test polypeptide sequences for screening are represented. The plurality of different short peptide sequences can be randomly generated by any method (e.g. NNK or NNN nucleotide randomization). The plurality of different short peptide sequences can also be preselected, either by previous experiments selecting for binding to a target, or from existing data sets in the scientific literature that have reported rationally-designed peptide libraries.

(118) The host cell can additionally be made to be permeable to small molecules, for example by deletion of drug efflux pumps, such as PDR5, ERG6, or 12gene Δ HSR (*Chinen*, 2011), to enable a small molecule screening approach.

(119) The host cell can additionally carry mutations to enable more efficient transformation with vectors and/or more efficient uptake small molecules.

(120) PLASMIDS 1, 2, and 3 can be used in various permutations. In some embodiments, integration of PLASMID 1 into the genome of the host cell (as confirmed using PLASMID 3) is followed by transformation of a library of PLASMID 2 with randomly encoded peptides using, for example, NNK or NNN codons.

(121) In some embodiments, to perform a screen to identify a peptide that can disrupt a PPI, the host cell is propagated in selection media to ensure the presence of PLASMID 1 and a proper PPI (e.g. on media lacking the positive selection marker for yeast, or in media containing antibiotic for human or bacterial cells). This host cell is then be transformed with PLASMID 2, and immediately

transferred to selection media to ensure all components are present (i.e. on media lacking both plasmid selection markers for yeast, or antibiotics for bacterial or mammalian cells), and are inducing expression of any inducible component (e.g. with Gal, doxycycline, etc).

(122) In other embodiments, the plasmids are used as a 'plug and play platform' utilizing the yeast mating type system, where the 2 or more plasmids (or the genetic elements therein) are introduced into the same cell by cell fusion or cell fusion followed by meiosis instead of transfection. This cell fusion involves two different yeast host cells bearing different genetic elements. In this embodiment, yeast host cell 1 is one of MATa or MATalpha and includes an integration of PLASMID 1. In this embodiment, yeast host cell 1 strain can be propagated on positive selection media to ensure a proper PPI is present. In this embodiment, the yeast host cell 2 can be the opposite mating type. This strain carries (or has integrated) the randomized peptide library and 'death agent' (e.g. cytotoxic reporter) plasmid (PLASMID 2). Yeast host cell 2 can be generated via large batch high efficiency transformation protocols which ensure a highly diversified library variation within the cell culture. Aliquots of this library batch can then be frozen to maintain consistency. In this embodiment, the strains are mated in batch to result in a diploid strain that carries all the markers, the PPI, positive selection, 'death agents' and peptide. This batch culture then can be propagated on solid medium that enables selection of all the system components (i.e. media lacking both positive selection markers), and inducing expression of any inducible component (i.e. with Gal).

(123) Surviving colonies from limiting dilution experiments performed on host cells bearing both the Y2/3H and library/cytotoxic constructs (either introduced to the cell by transfection or mating) can constitute colonies with a specific PPI that has been disrupted by a peptide and no longer triggers the death cascade triggered by the encoded 'death agents' (e.g. cytotoxic reporters) while maintaining a differential PPI driving a positive selection marker. The peptide sequence can be obtained by DNA sequencing the peptide-encoding region of PLASMID 2 in each surviving colony.

(124) To ensure that survival is due to inhibition of the PPI rather than stochastic chance or faulty gene expression, an inducible promoter can be used to inactivate the production of either the PPI or the peptide and confirm specificity. In some embodiments, cell survival is observed only on media with galactose wherein all the components are expressed; and no survival is observed on media without galactose when expression of the peptide is lost.

(125) The plasmids can also be isolated and re-transformed into a fresh host cell to confirm specificity. Biochemical fractionation of the viable host cells which contain the PPI, peptide, positive selection and 'death agent' followed by pull-down experiments can confirm an interaction between the peptide sequence and either PPI partner using the encoded tags (e.g. Myc-tag, HA-tag, His-tag). This is also helpful to perform SAR to determine the binding interface.

(126) The peptides to be used in screening assay can be derived from a complex library that involves post-translational modifying enzymes. The modified peptides can be analyzed by methods such as mass spectrometry, in addition being sequenced to ID the primary sequence. The peptides can also be tested for inherent membrane permeability by reapplying them onto the host cells exogenously (from a lysate) and observing for reporter inactivation or activation.

(127) Once enough surviving host cell colonies are sequenced, highly conserved sequence patterns can emerge and can be readily identified using a multiple-sequence alignment. Any such pattern can be used to 'anchor' residues within the library peptide insert sequence and permute the variable residues to generate diversity and achieve tighter binding. In some embodiments, this can also be done using an algorithm developed for pattern recognition and library design. Upon convergence, the disrupting peptide pattern, as identified through sequencing, can be used to define a peptide disruptor sequence. Convergence is defined by the lack of retrieval of any new sequences in the last iteration relative to the penultimate one.

EXAMPLES

Example 1: Method for Identifying a Disruptor of a Protein-Protein Interaction

(128) This is an example of a system that uses two variants of one protein, fused to different DBDs to identify inhibitors against a specific PPI. The PPI integration plasmid (PLASMID 1; FIG. 8) is used to integrate into *Saccharomyces cerevisiae* three proteins that constitute the binary PPIs of interest. The plasmid encodes for the fusion of an AD (VP16) with c-Raf, a DBD (LexA) with KRas(G12D), and a DBD (Gal4BD) with KRas, driven by a strong promoter and terminator ADH1. Fusion proteins VP16-c-Raf, LexA-KRas(G12D), and Gal4BD-KRas are tagged with MYC, FLAG, and HA, respectively. The plasmid further includes bacterial selection and propagation markers (ori and AmpR), and yeast replication and selection markers (TRP1 and CEN). The plasmid also has sites for integration into the genome at a specified locus.

(129) *Saccharomyces cerevisiae* is co-transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library (NNK 20-mer sequences). The plasmid is driven by a strong promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence of methionine-valine-asparagine (MVN) to maximize the half-life of the peptide, and terminated with an untranslated region (UTR) of a short protein (sORF1). The selection and library plasmid also comprises a sequence that encodes a His-tag. The translated protein has the N-terminus of methionine to minimize proteolysis.

(130) The selection and library plasmid additionally comprises a LexAop sequence, which induces 'death agents' (cytotoxic reporter expression) when bound by a functional transcriptional factor that is formed by LexA-KRas(G12D) fusion protein and VP64-c-Raf fusion protein, unless interrupted by a disrupter peptide. The selection and library plasmid also contains a positive selection marker, ADE2 which is under control of Gal4BD-KRas. The plasmid further includes bacterial selection and propagation markers (ori and AmpR), and yeast replication and selection markers (TRP1 and CEN).

(131) To confirm expression of the reporters and the successful construction of the strains, *Saccharomyces cerevisiae* is transformed with a confirmation plasmid (PLASMID 3; FIG. 10). The confirmation plasmid includes a direct fusion between the AD fusion protein (VP64-c-Raf) and DBD fusion proteins (LexA-KRas(G12D) or Gal4BD-KRas). The confirmation plasmid includes yeast replication and selection markers (TRP1 and CEN). The confirmation plasmid is used to confirm integration of LexAop promoter before the ADE2 gene. Transformation with the confirmation plasmid allows for the activation of ADE2 only if the promoter is properly integrated. Also, confirmation of death agents is enabled by inducible expression of the confirmation fusion construct.

(132) The screen is performed by mating the strains in a batch to result in a diploid strain, which carries all the markers, the PPIs, the positive selection, the death agents, and the peptide. This batch culture is then propagated on solid medium, which enables selection of all the system components (media lacking two nutritional components) and induces expression of any inducible component with Gal.

(133) Surviving colonies constitute colonies with a specific PPI(KRas-c-Raf) that have been disrupted by a peptide and no longer trigger the death cascade induced by the encoded death agents and maintain positive selection of the remaining PPI.

(134) The peptide sequence that disrupts the death agent-driving PPI is obtained by DNA sequencing the peptide-encoding region of the selection and library plasmid in each surviving colony.

(135) To confirm specificity, the inducible marker is used to inactivate the production of the PPI and confirm specificity. The plasmid is then isolated and re-transformed into a fresh parental strain to confirm specificity.

(136) Biochemical fractionation of the viable strain that contained the PPI, peptide, selection marker, and death agent, followed by pull-down experiments is done to confirm an interaction between the peptide sequence and either PPI partner using the encoded tags.

(137) An alternative example can be made by switching LexA with TetR. In another alternative example, fusion proteins in the PPI integration plasmid and the randomized peptide library in selection and library plasmid are driven by an inducible promoter, GAL1, instead of ADH1 used alternatively in another example. In yet another example, the N-terminus of the peptide translated from the selection and library plasmid can alternatively be glycine, alanine, serine, threonine, valine, or proline. In other examples, the genetic reporter in the confirmation plasmid is HIS3 or URA3, in place of ADE2. Either mating types of *Saccharomyces cerevisiae* haploid state can be used as background strain in alternative examples. In other examples, background strains also express the enzymes for the cyclization and methylation of peptides like lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP), patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and GmPOPB from *Galerina marginata* or other species.

Example 2: Method for Identifying Protein-Protein Interaction Disruptor

(138) This is an example of system that uses two different proteins, fused to different DBDs to identify inhibitors against a specific PPI. The PPI integration plasmid (PLASMID 1; FIG. 8) is used to integrate into *Saccharomyces cerevisiae* three proteins that constitute the binary PPIs of interest. The plasmid encode for the fusion of an AD (VP64) with TEAD, a DBD (LexA) with YAP, and a DBD (Gal4BD) with VGLL4, each associated with ADH1 promoter. Three protein fusion sequences are tagged with either FLAG, MYC or HA. The plasmid further includes yeast replication and selection markers (TRP1 and CEN). The plasmid also has sites for integration into the genome at a specified locus.

(139) The *Saccharomyces cerevisiae* is co-transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library (NNK 20-mer sequences). The selection plasmid is driven by a strong promoter, ADH1. The selection and library plasmid also comprises a sequence that encodes a His-tag.

(140) The selection and library plasmid additionally comprises a LexAop sequence, which induces 'death agent' (cytotoxic reporter expression) when bound by a functional transcriptional factor that is formed by LexA-YAP fusion protein and VP64-TEAD fusion protein, unless interrupted by a disrupter peptide. The selection and library plasmid also contains a positive selection marker, ADE2 which is under control of Gal4BD-VGLL4 and VP64-TEAD. The plasmid further includes yeast replication and selection markers (TRP1 and CEN).

(141) To confirm expression of the reporters and the successful construction of the strains, *Saccharomyces cerevisiae* is transformed with a confirmation plasmid (PLASMID 3; FIG. 10). The confirmation plasmid includes a direct fusion between the AD fusion protein (VP64-TEAD) and DBD fusion proteins (LexA-YAP or Gal4BD-VGLL4). The confirmation plasmid includes yeast replication and selection markers (TRP1 and CEN). The confirmation plasmid is used to confirm integration of LexAop promoter before the ADE2 gene. Transformation with the confirmation plasmid allows for the activation of ADE2 only if the promoter is properly integrated. Also, confirmation of death agents is enabled by inducible expression of the confirmation fusion construct.

(142) The screen is performed by propagating the parental strain on selection media to ensure the presence of the PPI Integration plasmid, and that a proper PPI has occurred, which is confirmed via use of the confirmation plasmid. The strain is cultured on media lacking nutrient markers against positive selection markers to ensure selection of colonies where the desired interaction occurred. The strain is then transformed with the selection and library plasmid, and is immediately plated on selection media to ensure all components are present (on media lacking the two nutritional markers) and is induced expression of any inducible component (with Gal).

(143) Surviving colonies constitute colonies with a specific PPI(YAP-TEAD) that has been disrupted by a peptide and no longer triggers the death cascade induced by the encoded death agents and maintain positive selection of the remaining PPI.

(144) The peptide sequence that disrupts the death agent-driving PPI is obtained by DNA sequencing the peptide-encoding region of the selection and library plasmid in each surviving colony.

(145) To confirm specificity, the inducible marker is used to inactivate the production of the PPI and confirm specificity. The plasmid is then isolated and re-transformed into a fresh parental strain to confirm specificity.

(146) Biochemical fractionation of the viable strain that contained the PPI, peptide, selection marker, and death agent is followed by pull-down experiments to confirm an interaction between the peptide sequence and either PPI partner using the encoded tags.

(147) An alternative example can be made by switching LexA with TetR. In another alternative example, fusion proteins in the PPI integration plasmid and the randomized peptide library in selection and library plasmid are driven by an inducible promoter, GAL1, instead of ADH1 used alternatively in another example. In yet another example, the N-terminus of the peptide translated from the selection and library plasmid can alternatively be glycine, alanine, serine, threonine, valine, or proline. In other examples, the genetic reporter in the confirmation plasmid is HIS3 or URA3, in place of ADE2. Either mating types of *Saccharomyces cerevisiae* haploid state can be used as background strain in alternative examples. In other examples, background strains also express the enzymes for the cyclization and methylation of peptides like lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP), patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and GmPOPB from *Galerina marginata* or other species.

Example 3: Method for Identifying Protein-Protein Interaction Facilitator

(148) This is an example of system that uses two variants of one protein, fused to different DBDs to identify facilitator for a specific PPI. The PPI integration plasmid (PLASMID 1; FIG. 8) is used to integrate into *Saccharomyces cerevisiae* three proteins that constitute the binary PPIs of interest. The plasmid encodes for the fusion of an AD (VP64) with Mdm2, a DBDs (LexA) with KRas, and a DBD (Gal4BD) with KRas(G12D). DBD-KRas, each driven by ADH1 promoter. Three protein fusion sequences are tagged with either FLAG, MYC or HA. The plasmid further includes yeast replication and selection markers (TRP1 and CEN). The plasmid also has sites for integration into the genome at a specified locus.

(149) The *Saccharomyces cerevisiae* is co-transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 20-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The selection and library plasmid also comprises a sequence that encodes a HIS tag.

(150) The selection and library plasmid additionally comprises a LexAop sequence, which induces 'death agents' (cytotoxic reporter expression) when bound by a functional transcriptional factor that is formed by LexA-KRas fusion protein and VP64-Mdm2 fusion protein, when mediated by a facilitator peptide. The selection and library plasmid also contains a positive selection marker, ADE2 which is under control of Gal4BD-KRas(G12D) fusion protein and VP64-Mdm2 fusion protein and leading to expression of the positive selection marker when the fusion proteins are mediated by a facilitator. The plasmid further includes yeast replication and selection markers (TRP1 and CEN).

(151) To confirm expression of the reporters and the successful construction of the strains, *Saccharomyces cerevisiae* is transformed with confirmation plasmid (PLASMID 3; FIG. 10). The confirmation plasmid includes a direct fusion between the AD fusion protein (VP64-Mdm2) and DBD fusion proteins (LexA-KRas or Gal4BD-KRas(G12D)). The confirmation plasmid includes yeast replication and selection markers (TRP1 and CEN). The confirmation plasmid is used to confirm integration of LexAop promoter before the ADE2 gene. Transformation with the confirmation plasmid allows for the activation of ADE2 only if the promoter is properly integrated. Also, confirmation of death agents is enabled by inducible expression of the confirmation fusion

construct.

(152) The screen is performed by mating the strains in a batch to result in a diploid strain, which carries all the markers, the PPIs, the positive selection, the death agents, and the peptide. This batch culture is then propagated on solid medium, which enable selection of all the system components (media lacking two nutritional components) and induce expression of any inducible component with Gal.

(153) Surviving colonies constitute colonies with a specific PPI(KRas(G12D)-Mdm2) that has been facilitated by a peptide and do not have nonspecific PPI(KRas-Mdm2) that can trigger the death cascade induced by the encoded death agents and maintain positive selection of the remaining PPI.

(154) The peptide sequence that is able to facilitate a specific PPI is obtained by DNA sequencing the peptide-encoding region of the selection and library plasmid in each surviving colony.

(155) To confirm specificity, the inducible marker is used to inactivate the production of the PPI and confirm specificity. The plasmid is then isolated and re-transformed into a fresh parental strain to confirm specificity.

(156) Biochemical fractionation of the viable strain that contained the PPI, peptide, selection marker, and death agent is followed by pull-down experiments to confirm an interaction between the peptide sequence and either PPI partner using the encoded tags.

(157) An alternative example can be made by switching LexA with TetR. In another alternative example, fusion proteins in the PPI integration plasmid and the randomized peptide library in selection and library plasmid are driven by an inducible promoter, GAL1, instead of ADH1 used alternatively in another example. In yet another example, the N-terminus of the peptide translated from the selection and library plasmid can alternatively be glycine, alanine, serine, threonine, valine, or proline. In other examples, the genetic reporter in the confirmation plasmid is HIS3 or URA3, in place of ADE2. Either mating types of *Saccharomyces cerevisiae* haploid state can be used as background strain in alternative examples. In other examples, background strains also express the enzymes for the cyclization and methylation of peptides like lanthipeptides maturation enzymes from *Lactococcus lactis* (LanB, LanC, LanM, LanP), patellamide biosynthesis factors from cyanobacteria (PatD, PatG), butelase 1 from *Clitoria ternatea*, and GmPOPB from *Galerina marginata* or other species.

Example 4: Reversible Induction of a Nutritional Reporter by Protein-Protein Interaction

(158) This is an example of two platforms that either used two variants or two different proteins, fused to different DBDs to identify inhibitor by nutrient based selection. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in *Saccharomyces cerevisiae* cells using an integration plasmid (FIG. 6). In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintained their interaction to drive expression of Nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^{sup.4} cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30° C. The results showed that the cells grown on media that selected for Nutritional reporter 2 had poor survival rate when added the inhibitor, illustrating the specificity of the inhibitor for KRas(G12D) and c-Raf interaction.

(159) In the second platform, VGLL4 or YAP fused to DBD and TEAD fused to AD were expressed in *Saccharomyces cerevisiae* cells with an integration plasmid (FIG. 6). In the absence of any inhibitors, the DBD fusion protein and AD fusion protein pairs maintained their interaction to drive expression of Nutritional reporters 1 and 2. A 5-fold dilution series starting at 10^{sup.4} cells were spotted onto selective media with or without inhibitor and visualized after 2 days of growth at 30 C. The results showed that the cells grown on media that selected for Nutritional reporter 2 had particularly poor survival rate when added the inhibitor, illustrating the specificity of the inhibitor for YAP and TEAD.

Example 5: Reversible Induction of a Cytotoxic Reporter by Protein-Protein Interaction

(160) This is an example of two platforms that used either two variants or two different proteins,

fused to different DBDs to identify inhibitors by toxicity based selection. In the first platform, KRas and KRas(G12D) fused to DBDs and c-Raf fused to AD were expressed in *Saccharomyces cerevisiae* cells with an integration plasmid. In the absence of inhibitors, the KRas(G12D) and c-Raf maintained an interaction to drive expression of cytotoxic reporter. A nutritional reporter was controlled by KRas and c-Raf interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30° C. In cell populations with KRas(G12D) and c-Raf interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the inhibitor to KRas(G12D) and c-Raf interaction. In the second platform, VGLL4 or YAP fused to DBD and TEAD fused to AD were expressed in *Saccharomyces cerevisiae* cells with an integration plasmid. In the absence of inhibitors, the YAP and TEAD maintained their interaction to drive expression of cytotoxic reporter. A nutritional reporter was controlled by VGLL4 and TEAD interaction. The cells were patched onto selective media for a nutritional marker with or without inhibitor and visualized after 4 days of growth at 30 C. In cell populations with YAP and TEAD interaction, only those with the inhibitor showed enhanced cell viability, illustrating the specificity of the inhibitor to YAP and TEAD interaction.

Example 6: Cyclization of Peptides

(161) This is an example of a system that induces cyclization of randomized peptides in a complex to achieve enhanced cell permeability and peptide stability. The *Saccharomyces cerevisiae* is transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 60-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence that encodes Methionine Valine Asparagine (MVN) at N-terminus of peptide to maximize the half-life of the peptide, and terminate with sequence that encodes the UTR of a short protein (sORF1). The peptide is also tagged with a HIS tag.

(162) The variable peptide library region of the selection and library plasmid is embedded within primary sequence of modifying enzyme, a homolog of N-methyltransferase from *Rhizophogun vinicolor* and contains randomized residues. The diversified variable region is excised and end-to-end cyclized by the action of a beta-lactamase DD-transpeptidase from *R. vinicolor*. Some of the side chains of the randomized residues are subsequently post-translationally isomerized from L- to D-configuration and hydroxylated.

Example 7: Cyclization of Peptides

(163) This is an example of a system that induces cyclization of randomized peptides in a complex to achieve enhanced cell permeability and peptide stability. The *Saccharomyces cerevisiae* is transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 60-mer sequences. The selection plasmid is driven by a strong promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence that encodes Methionine Valine Asparagine (MVN) at N-terminus of peptide to maximize the half-life of the peptide, and terminate with sequence that encodes the UTR of a short protein (sORF1). The peptide is also tagged with a HIS tag.

(164) The variable peptide library region of the selection and library plasmid is embedded within primary sequence of modifying enzyme, a homolog of prolyl endopeptidases belonging to the PopB family and contains randomized residues. A post-translational processing of the variable peptide by the co-expressed prolyl endopeptidases leads to the generation of N-to-C cyclized macrocycles.

Example 8: Bicyclization of Peptides

(165) This is an example of a system that induces cyclization of randomized peptides in a complex to achieve enhanced cell permeability and peptide stability. The *Saccharomyces cerevisiae* is transformed with the selection and library plasmid (PLASMID 2; FIG. 9) for the expression of a randomized peptide library, NNK 60-mer sequences. The selection plasmid is driven by a strong

promoter, ADH1. The initiation sequence of the selection and library plasmid is a fixed sequence that encodes Methionine Valine Asparagine (MVN) at N-terminus of peptide to maximize the half-life of the peptide, and terminated with sequence that encodes the UTR of a short protein (sORF1). The peptide is also tagged with a HIS tag.

(166) The variable peptide library region of the selection and library plasmid is embedded within primary sequence of modifying enzyme, a homolog of prolyl endopeptidases belonging to the PopB family and contains randomized residues. A post-translational processing of the variable peptide by the co-expressed prolyl endopeptidases leads to the generation of N-to-C cyclized macrocycles.

(167) The macrocycles is then hydroxylated at the 2-position of the indole ring of the tryptophan residue by a hydroxylase belonging to the Cytochrome P450 family of oxygenases. A condensation reaction is followed catalyzed by dehydratase to form a tryptathionine bridge between the 2'-hydroxyl position on tryptophan and the thiol group from the cysteine residue. A flavin-monoxygenase enzyme converts the intermediate product to a sulfinyl form by S-oxygenation of the tryptathionine thiol. Some of the side chains of the peptide are subsequently hydroxylated at position 6 on the indole ring of the tryptathionine-forming tryptophan residue by a P450 family monoxygenase. The resulting bicyclized macrocycles comprises a tryptathionine bridge.

(168) While preferred embodiments of the present disclosure have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the disclosure. It should be understood that various alternatives to the embodiments of the disclosure described herein may be employed in practicing the disclosure.

Claims

1. A method of screening for a molecule that disrupts an interaction between a first test protein and a second test protein in a host cell, while not disrupting an interaction between a third test protein and a second test protein in the host cell, the method comprising: expressing in the host cell a first fusion protein comprising the first test protein and a first DNA-binding moiety; expressing in the host cell a second fusion protein comprising the second test protein and a gene activating moiety; expressing in the host cell a third fusion protein comprising a third test protein and a second DNA-binding moiety, wherein the second DNA-binding moiety is different from the first DNA-binding moiety; and delivering a molecule to the host cell to test for disruption of the interaction of the first test protein and the second test protein without disrupting the interaction between the third test protein and the second protein; wherein a plurality of death agent genes are disposed within the host cell, each operably linked to a promoter DNA sequence specific for the first DNA-binding moiety, wherein a positive selection reporter is disposed within the host cell and operably linked to a promoter DNA sequence specific for the second DNA binding moiety, wherein, when the molecule does not disrupt the interaction between the first test protein and the second test protein and does not disrupt the interaction between the second test protein and the third test protein, an interaction between the first test protein and the second test protein causes the gene activating moiety to activate expression of the death agent genes, while an interaction between the second test protein and the third test protein causes the gene activating moiety to activate expression of the positive selection reporter; wherein, when the molecule disrupts the interaction between the first test protein and the second test protein but not the third test protein and the second test protein, (1) an interaction between the second test protein and the third test protein causes the gene activating moiety to activate expression of the positive selection reporter; and (2) disruption of the interaction between the first test protein and the second test moiety does not activate expression of the death agents, and wherein the host cell is an animal cell.

2. The method of claim 1, wherein the host cell comprises an integrated DNA encoding the first

fusion protein, an integrated DNA encoding the second fusion protein, an integrated DNA encoding the third fusion protein, a plasmid DNA encoding death agents produced by the death agent genes, and a plasmid DNA encoding the positive selection reporter.

3. The method of claim 1, wherein the first test protein is a Ras variant, the second test protein is a Raf kinase, and the third test protein is Ras.

4. The method of claim 1, wherein the first DNA-binding moiety, the second DNA-binding moiety, or both, is a LexA, cI, Gli-1, YY1, Glucocorticoid receptor, TetR, or Ume6 DNA binding moiety.

5. The method of claim 1, wherein the gene activating moiety is a VP16, GAL4, NF- κ B, B42, BP64, VP64, or p65 gene activating moiety.

6. The method of claim 1, wherein death agent(s) encoded by the death agent genes are selected from the group consisting of: a ribosomally encoded xenobiotic agent, a ribosomally encoded poison, a ribosomally encoded endogenous or exogenous gene that results in severe growth defects upon mild overexpression, a ribosomally encoded recombinase that excises an essential gene for viability, a limiting factor involved in the synthesis of a toxic secondary metabolite, and any combination thereof.

7. The method of claim 1, wherein death agent(s) encoded by the death agent genes are selected from the group consisting of: Cholera toxin, SpvB toxin, CARDS toxin, SpyA Toxin, HopU1, Chelt toxin, Certhrax toxin, EFV toxin, ExoT, CdtB, Diphtheria toxin, ExoU/VipB, HopPtoE, HopPtoF, HopPtoG, VopF, YopJ, AvrPtoB, SdbA, SidG, VpdA, Lpg0969, Lpg1978, YopE, SptP, SopE2, SopB/SigD, SipA, YpkA, YopM, Amatoxin, Phalloidin, Killer toxin KP1, Killer toxin KP6, Killer Toxin K1, Killer Toxin K28 (KHR), Killer Toxin K28 (KHS), Anthrax lethal factor endopeptidase, Shiga Toxin, Saporin Toxin, Ricin Toxin, and any combination thereof.

8. The method of claim 1, wherein the molecule is a small molecule.

9. The method of claim 8, wherein the small molecule is a peptidomimetic.

10. The method of claim 1, wherein the molecule is a peptide or a protein.

11. The method of claim 10, wherein the peptide or the protein is a naturally occurring protein product, is a synthesized protein product, or is produced by expression of a recombinant gene.

12. The method of claim 10, wherein the peptide or protein is expressed from a library comprising test DNA molecules comprising DNA sequences that encode polypeptides, and wherein the peptide or the protein is expressed from the test DNA molecules.

13. The method of claim 12, wherein the polypeptides are 60 or fewer amino acids in length.

14. The method of claim 12, wherein the polypeptides are processed into cyclic or bicyclic peptides in the host cell.

15. The method of claim 1, wherein the first test protein is YAP or TAZ, the second test protein is TEAD, and the third test protein is VGLL4.

16. The method of claim 1, wherein the animal cell is a mammalian cell.
